

TEAL PDU Module (5140621) in XGD Cabinet Operations Manual



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Revision 3
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Language Policy

Direction 2128126 - Language Policy For Service Documentation

ПРЕДУПРЕЖ ДЕНИЕ (BG)	Това упътване за работа е налично само на английски език. - Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод. - Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа. - Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.
警告 (ZH-CN)	本维修手册仅提供英文版本。 - 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。 - 未详细阅读和完全理解本维修手册之前，不得进行维修。 - 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。
警告 (ZH-HK)	本服務手冊僅提供英文版本。 - 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。 - 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。 - 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。
警告 (ZH-TW)	本維修手冊僅有英文版。 - 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。 - 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。 - 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。
UPOZOR- ENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. - Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.

VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. • V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. • Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. • V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. • Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. • Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. • Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.
WAARSCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. • Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. • Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. • Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.
WARNING (EN)	This service manual is available in English only. • If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services. • Do not attempt to service the equipment unless this service manual has been consulted and is understood. • Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles. • Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. • Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. • Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. • Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. • Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. • Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. • Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. • Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. • Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. • Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. • Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. • Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο. • Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης. • Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις. • Εάν δεν λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. • Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése. • Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. • Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramlítás, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fánleg á ensku. - Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu. - Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. - Brot á sinna þessari aðvörðun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklinga frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. - Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. - Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. - Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.
警告 (JA)	このサービスマニュアルには英語版しかありません。 - サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 - このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 - この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다. - 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. - 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. - 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.
BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. - Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. - Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. - Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS (LT)	Šis eksploataavimo vadovas yra tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. • Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploataavimo vadovo. • Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. • Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.
ATENÇÃO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. • Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖН О! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZOR- ENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. • Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. • Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. • Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNE- NIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCIÓN (ES)	Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. • Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
OPOZORILO (SL)	Ta servisni priročnik je na voljo samo v angleškem jeziku. • Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod. • Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli. • Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. • Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. • Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. • Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.
ЗАСТЕРЕЖЕННЯ (UK)	Даний посібник з експлуатації доступний тільки англійською мовою. • Якщо постачальник послуг клієнта спілкується іноземною мовою, тоді клієнт зобов'язаний забезпечити переклад. • Заборонено проводити огляд обладнання без попереднього звертання до даного посібника з експлуатації і розуміння інформації, поданої у ньому. • Недотримання цього застереження може завдати шкоди здоров'ю постачальника послуг, оператора або пацієнта через ураження електричним струмом, механічну травму або інше ушкодження.

Revision History

Revision	Date	Description
DOC1805613, Revision 3	July 2021	To adhere to publication standards, the number on the front page reflects the DOC number and will be used going forward. In Chapter 4, Field Replaceable Units, updated the following part numbers: 3-Pole Power Contactor for E-stop: from 5140621-3 to 5140621-11 48V Power Supplies Bracket: from 534114-10 to 5343114-10
Vendor Manual Revision 4	August 12, 2020	Routed in MyWorkshop with DOC1805613, Revision 2. - Document format updated to reflect SIMS standards. - A note was added to the Revision History for Revision 03: "Manual owned by vendor until Revision 03." - Added 3.1.1 Applying LOTO - MDP on page 7 and 3.1.2 Removing LOTO - MDP on page 11. 4.2.6 Replacement of power supplies on page 44 was updated to include information per <i>Service Note DOC2175961, 48V PDU Power Supply Replacement (Condor to Delta)</i>. This information was added as Appendix A - Replacement of power supplies. 4.1 Field Replaceable Units (FRUs) on page 19 updated to include supplementary information about maintenance and replacement, and to include additional parts in the FRU listing. - The following sections were added to the chapter "Field Replaceable Units." 4.2.7 Replacement of the main input circuit breaker (CB1) on page 44 4.2.8 Replacement of the gradient 420V power circuit breaker (CB2) on page 47 4.2.9 Replacement of superconducting shim power supply and magnet power supply circuit breakers (CB7 and CB8) on page 50 4.2.10 Replacement of BrainWave and gradient 48V circuit breakers (CB5 and CB9) on page 50 4.2.11 Replacement of receptacles and terminal blocks (REC1-2 and TB2) on page 51 4.2.12 Replacement of switches and test points (SW2 and TP1-7) on page 54 - In 4.2.13 Placement of new FRUs on page 55, step 1 was updated to include the new replacement sections.

(continued)		
Revision	Date	Description
		8. 4.2.14 Troubleshooting guide on page 56 was updated. 9. Appendix A - Replacement of power supplies was added.
Vendor Manual Revision 03	August 9, 2019	Manual owned by vendor until Revision 03. Revision 03 was not released; the changes below are included in the released Revision 4. Added wording and picture for removing contactors. Updated FRU part numbers. Updated instructions for PS bracket FRU. Updated TOC page numbers.
Vendor Manual Revision A2	June 7, 2012	- Added photos and explanation of control board switch configuration to section 4.1.3. - Added photo of LED power supply LED and explanation of LED purpose to section 4.1.6.
Vendor Manual Revision A1	September 22, 2008	As per customer request, update manual to include detailed procedure to replace FRUs. Changed spec to GE Specifications: 5140621PSP_r14
Vendor Manual revision A0	June 2, 2008	New Release.

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Important Safety Instructions

Save these instructions

This manual contains important instruction for the XGD Power Distribution Unit (PDU) that must be followed during installation, operation, and maintenance.



ELECTRIC SHOCK HAZARD

Opening enclosure exposes hazardous voltages.

Refer service to qualified personnel.

IMPORTANT

As standards, specifications, and designs change from time to time, please ask for confirmation of the information given in this publication.

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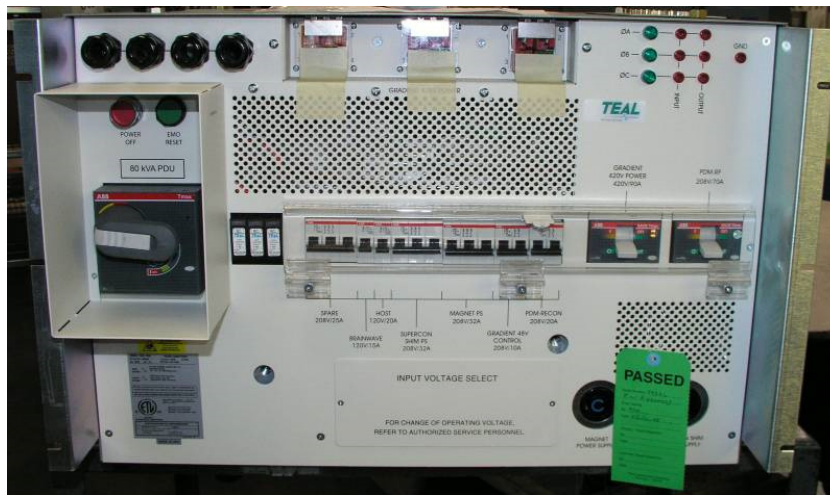
Chapter 1 Introduction

1.1 Introduction

1.1.1 General Description

The XGD Power Distribution Unit (PDU) distributes power to various components of a General Electric Medical Systems MR system while providing surge suppression and electrical isolation for the power line. Three phase input voltage is selectable among 380/400/415/480 VAC at a rated frequency of 50/60 Hz. The power rating for this PDU is 79 kVA continuous and 103 kVA instantaneous. The PDU mounts into the GE 5173027 cabinet.

Figure 1 XGD Power Distribution Unit



1.1.2 Scope

This manual describes procedures for installation and basic operation of the XGD Power Distribution Unit as well as procedures for failure diagnosis/isolation and installation of field replaceable components of the PDU.

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Chapter 2 Specifications

2.1 Specifications

GE Specifications: 5140621PSP

GE Part Number: 5140621

2.1.1 Electrical specifications

Power Rating: 79 kVA Continuous; 103 kVA Instantaneous

Input Voltage: 380/400/415/480 VAC Delta

Output Voltage: 208/120 VAC WYE and 242/420 VAC Wye

Output Circuit Breaker and Connection Requirements:

Breaker Size	Function	Ref. Desig	Sec Volt-age	Wire to Phase	E-Stop	Customer Access	Connec-tion Type
90A 3 Pole	Gradient 420V Pow- er	CB2	420	ABC	Yes	Front	APP#1321 G4, G5 (3 connectors)
70A 3 Pole	PDM – RF	CB3	208	ABC	Yes	Right Side	Breaker
25A 3 Pole	Spare	CB4	208	ABC	No	Top Left Side	Breaker
15A 1 Pole	Brainwave	CB5	120	A	No	Top Left Side	Breaker
20A 1 Pole	Host	CB6	120	A	No	Top Left Side	Breaker
32A 3 Pole	Supercon Shim PS	CB7	208	ABC	No	Front	5 wire twist lock (Nema style L21-30R Hubbell) (receptacle front of cab)

(continued)							
Breaker Size	Function	Ref. Desig	Sec Volt-age	Wire to Phase	E-Stop	Customer Access	Connec-tion Type
32A 3 Pole	Magnet Power Sup- ply	CB8	208	ABC	No	Front	5 wire twist lock (Nema style L21-30R Hubbell) (receptacle front of cab)
10A 2 Pole	Gradient 48V Control	CB9	208	BC	No	Top Right	7W2 D-Sub connectors
20A 2 Pole	PDM – Re- con	CB10	208	BC	No	Right Side	Breaker

Overload Protection: Input main circuit breaker (CB1) with adjustable trip settings. Output distribution circuit breakers.

Load Current:

83 Amps total maximum continuous (208V)

67 Amps total maximum continuous (420V), 103A Peak 5 sec maximum

Transformer Impedance: Less than 2%

Efficiency: 98% Typical

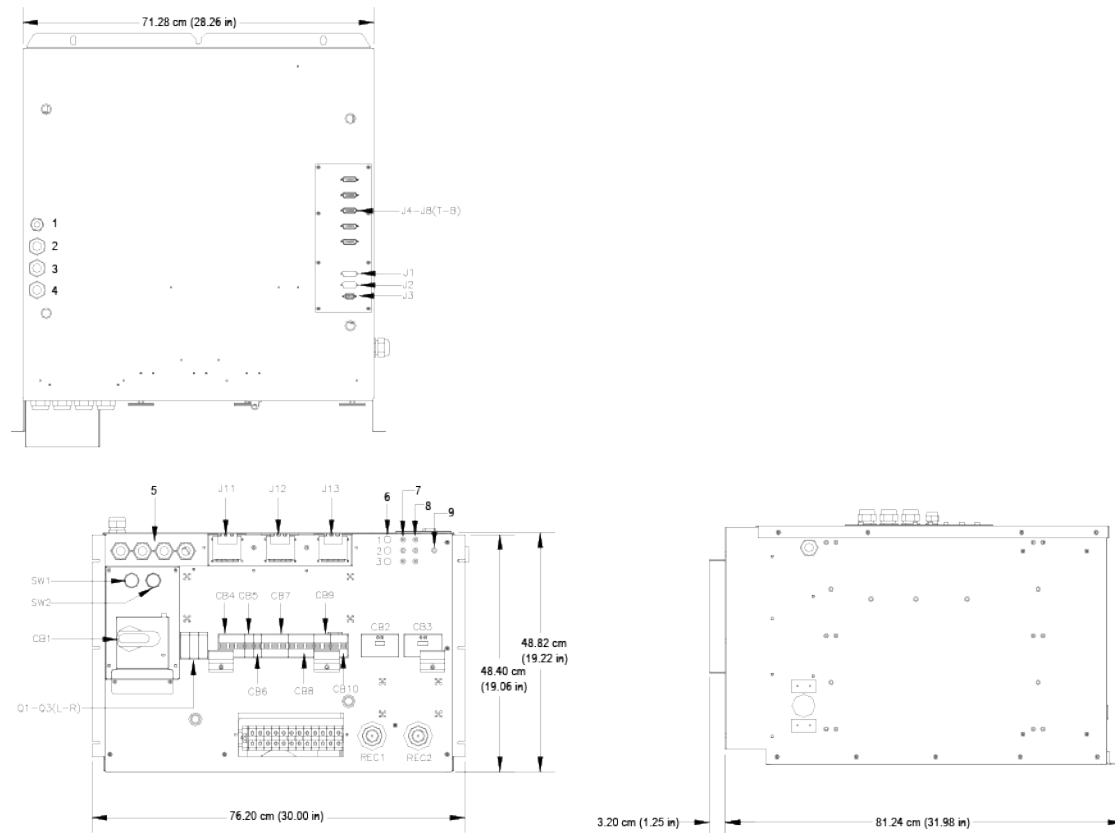
2.1.2 Environmental specifications

Heat Dissipation: 4,076 BTU/Hr

2.1.3 Mechanical specifications

Weight: 879 lbs / 398 kg

Dimensions:

Figure 2 XGD overall dimensions

1	Ground
2	Gradient 420V power
3	Spare
4	System cabinet
5	Main input
6	Indicator
7	Input LEDs
8	Output LEDs
9	Ground

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Chapter 3 Installation and operation

3.1 Lock out, tag out

3.1.1 Applying LOTO - MDP

1. Notify affected personnel working in the area where the LOTO is being performed.
2. Make sure the host computer is shut down.

NOTE

Powering off HEC without first powering down pumps from VFD can put the VFD at risk of damage.

3. Toggle off the pumps and blower with the PLC Signal Box keypad as follows:
 - Blower: Press and hold the up-arrow (^), and press **F1**
 - Gradient Coil Pump: Press and hold the up arrow (^), and press **F2**
 - Power Electronics: Press and hold the up arrow (^), and press **F3**

Figure 3 PLC signal box

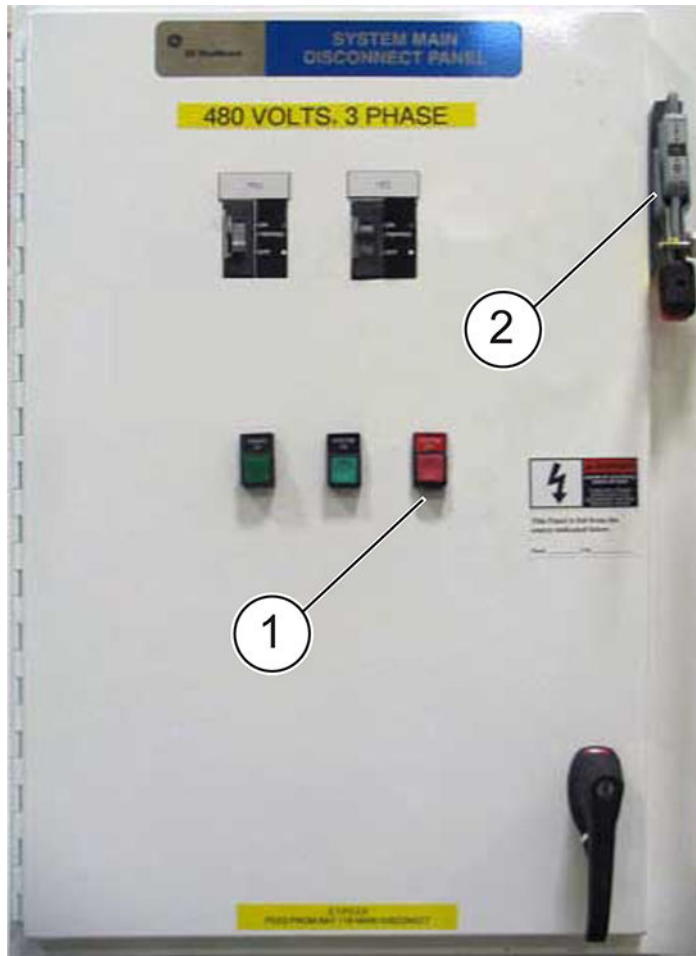


4. NOTE

Observe **Power On** light on the MDP. If the light is on, it signifies the main breaker is closed and control power is on.

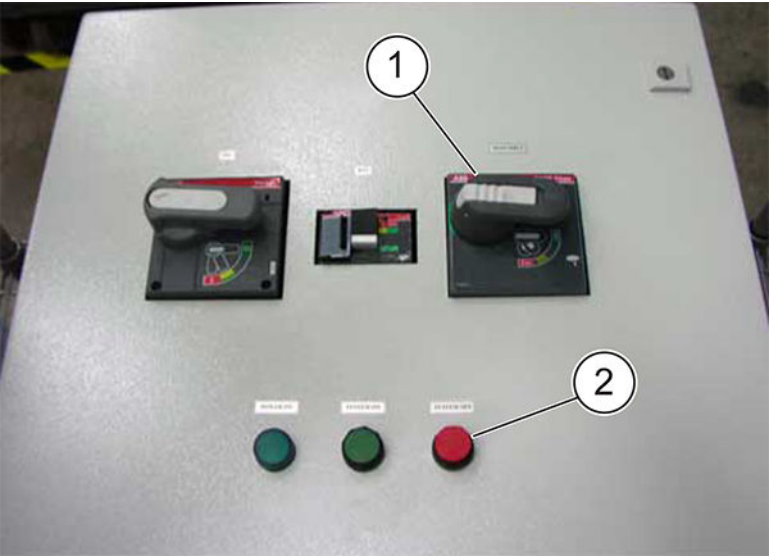
Press the **System Off** button on the MDP, which feeds the PGR cabinet and the HEC cabinet.

Figure 4 Main incoming switch on MDP with LOTO lock (GE MDP)



1	System off button
2	LOTO lock

Figure 5 Main incoming switch on MDP with LOTO lock (teal MDP)



1	LOTO lock
2	System off button

5. Turn the **Main Input Switch** (labeled with LOTO Lock) to the **OFF** position.
6. Left unchecked, the magnet monitor will call out reporting that the shield cooler compressor is powered off. Inhibit the reporting of false RFS events by placing the magnet monitor in Service mode.
7. Have a customer-supplied electrician disconnect the building main power from the MDP.
8. Apply your LOTO lock to the lockout/tagout device used for building main power.
9. **NOTE**
Observe **Power On** light on the MDP. The light should now be off. Do not assume that because the light is off that power is off. You must perform the next step to measure the voltage as a confirmation.

Open the MDP door and check that all voltage has dissipated. At the output of the main breaker, use a voltmeter to measure the voltage between each INPUT phase voltage test point and GND (ground). This reading should be <1V.

Figure 6 MDP test points (GE MDP)

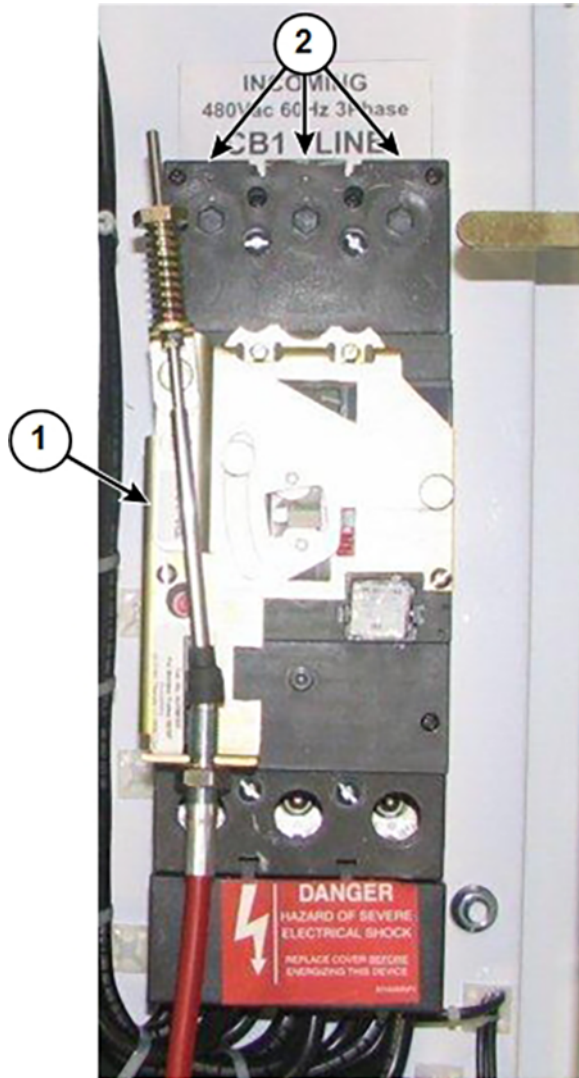
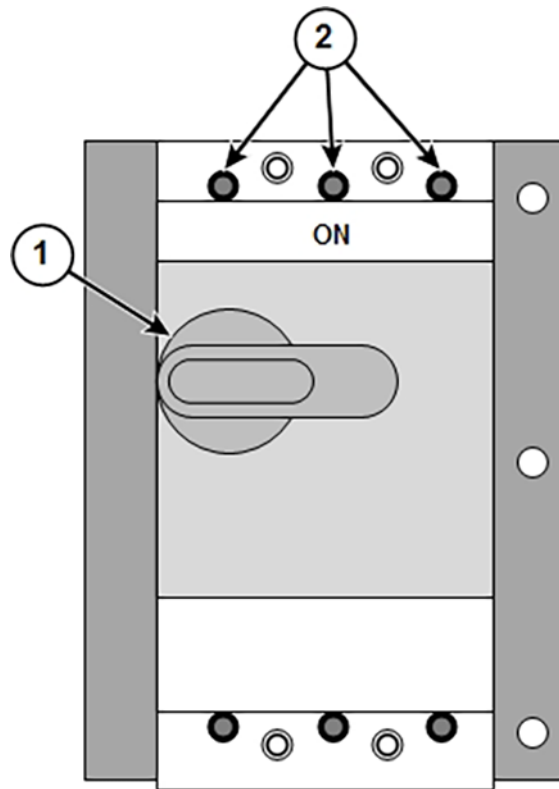


Figure 7 MDP test points (teal MDP)

10. Use a digital voltmeter to measure the voltage between each phase pair. This reading should be <1V.

3.1.2 Removing LOTO - MDP

1. Alert necessary personnel that power is being restored.
2. Remove the lock and tag.
3. Have a customer-supplied electrician reconnect building main power to the MDP.
4. On the MDP, check the breaker. If it is in a middle position, reset it by cycling to the Off position then back to the On position.
5. Switch on the Main Input Switch (labeled with LOTO lock).
6. On the MDP, push the green **System On** button.

7. On the PDU, press the **EMO RESET** button.

Figure 8 EMO RESET button



3.2 Installation

3.2.1 Input voltage selection

Before connecting the input power cables, determine the nominal value of the input voltage. The PDU allows for nominal input voltages of 380, 400, 415, or 480 VAC (three phase). Input voltage selection is accomplished as follows:

1. On the Main Disconnect Panel (MDP), do the steps in [3.1.1 Applying LOTO - MDP on page 7](#).
2. Using a voltmeter or other voltage indicating device at the Main Input Terminal Board, make sure that no voltage is being applied to the input of the PDU.
3. Remove the small cover panels from the front panel of the PDU marked INPUT VOLTAGE SELECT.
4. For each of the three phases, make sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.

Figure 9 Input voltage select terminal block (TB1)

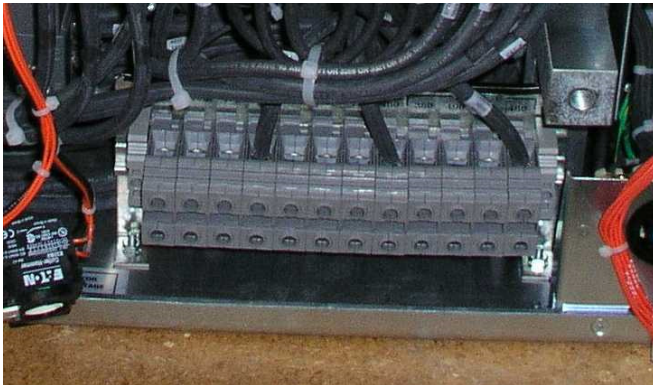
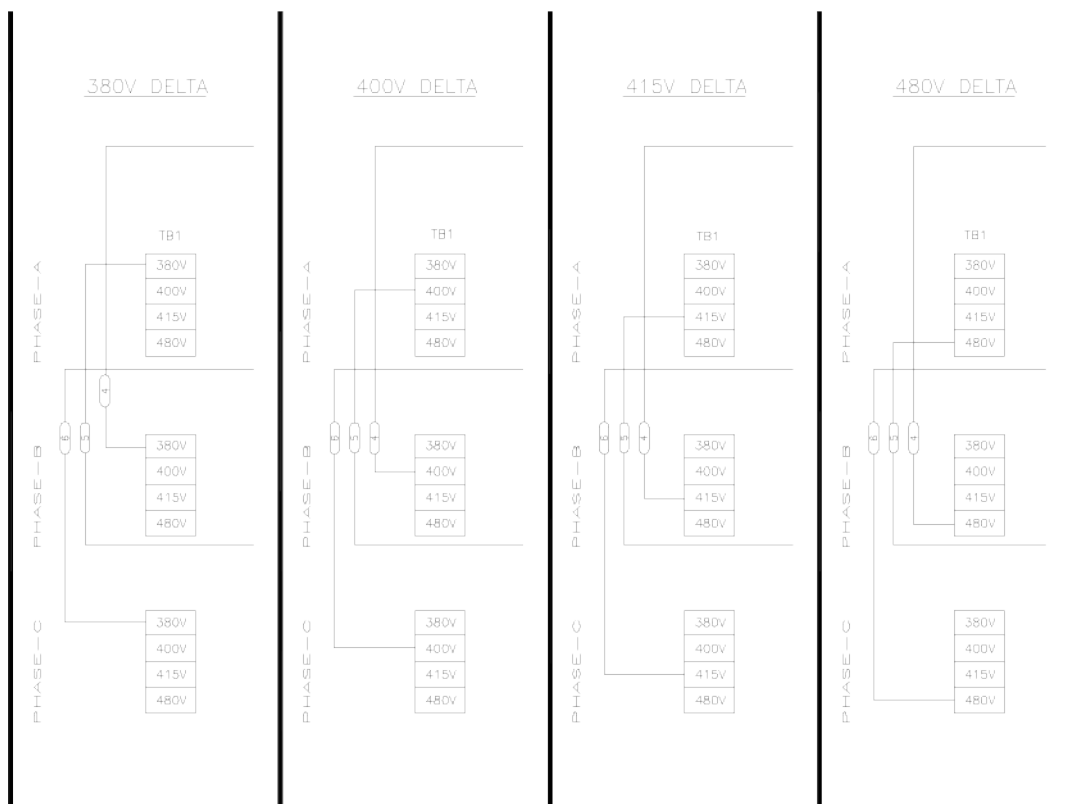


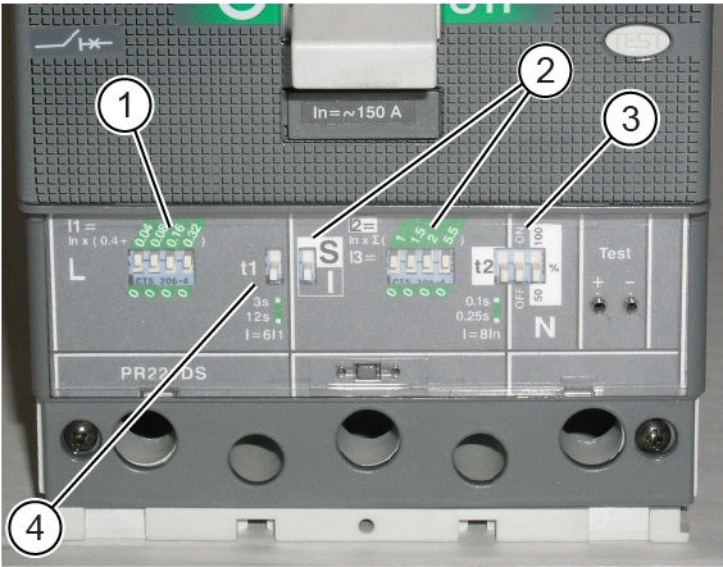
Figure 10 Input voltage selection

5. The overload and short circuit trip settings for the Input Circuit Breaker (CB1) must be set to the correct values corresponding to the input voltage. See [Maximum Input Current vs. Input Voltage](#).
 - a. Remove the small frame covering the main circuit breaker and the small panel underneath marked CURRENT ADJUST.
 - b. Open the transparent plastic cover on the lower portion of the circuit breaker to give access to the dip switches controlling the circuit breaker operation.
 - c. See [Circuit Breaker Dip Switch Settings](#) for the position of the dip switches for each input voltage selection. The “L” and “I” switches must match the configuration for the incoming voltage used as shown in [Input voltage selection](#).

Table 1 Maximum Input Current versus Input Voltage

Input Voltage	380V	400V	415V	480V
Maximum Input Current	121A	115A	111A	96A

Figure 11 Circuit breaker dip switch settings



1	Tripping threshold control switches (I1) Set as per Switch Settings for CBI table.
2	Instantaneous threshold value control switches (I3), (S/I) Set as per Switch Settings for CBI table.
3	t2 switches
4	Time-current curve control switch (t1)

Table 2 Switch settings for CBI

Input Voltage	380	400	415	480
CB AMP Size	150A	150A	140A	120A
CB "L" Settings	0.60	0.60	0.56	0.48
""L" Switch Settings	- - - -	- - - -	- - - -	- - - -
"t1" and "S/I" Set- tings	-	-	-	-
"I3" Switch Settings	- - - -	- - - -	- - - -	- - - -
"t2" Switch Set- tings	- - -	- - -	- - -	- - -

3.2.2 Connection

Note the Cable Access Panel shown in [Front Panel Component Locations](#) above the main circuit breaker. This panel provides strain relief cable clamps for the input and the various output cables. This panel may be removed from the PDU to provide easier access to the terminal blocks.

1. Route the cables through their respective cable clamps and connect cables to the appropriate terminals as shown.
2. Replace the Cable Access Panel and tighten the cable clamps.

IMPORTANT

For input cables, use wire sizes in accordance with National Electric Code (NEC) Table 310-16. This table gives maximum full-load input currents in amperes (which would occur at 10% undervoltage) for each of the input voltages, as a guide to selecting proper wire sizes.

3.3 Operation

3.3.1 Main input circuit breaker

The main input circuit breaker operating handle has three positions:

- OFF (as shown in [Figure 12 on page 16](#))
- ON (vertical), and
- TRIPPED (slightly to the right of vertical). If the main input circuit breaker has been tripped, it is necessary to move the handle to the OFF position to reset it before turning it ON.

Figure 12 Main input circuit breaker operating handle



The main input circuit breaker may be locked in the OFF position. By pressing on the small triangle on the operating handle, the lockout tab is made accessible to a padlock.

3.3.2 Power off

This pushbutton at the top of the front panel, when pressed, will immediately trip the main input circuit breaker, interrupting all power to the load.

3.3.3 EMO reset

This pushbutton, immediately to the right of the **Power Off** pushbutton, will enable the emergency stop circuit.

3.3.4 Transient Voltage Spike Suppressor (TVSS) indicators

To the right of the main input circuit breaker are three TVSS. Any module that fails must be replaced to restore the PDU to operation with surge protection functioning. See [4.2.1 Replacement of the TVSS module on page 22](#), for instructions on replacing a TVSS module.

3.3.5 Phase indicators

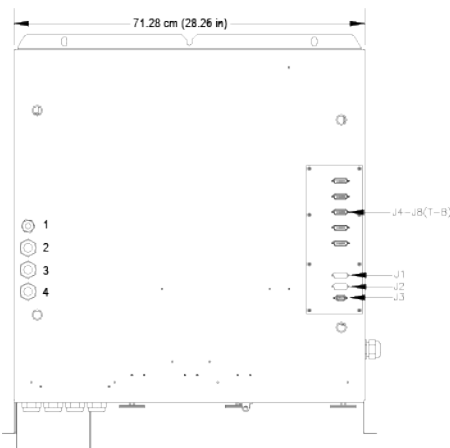
There are three phase indicator lights. These indicate that the transformer is energized and ready to deliver power to the load.

3.3.6 Input/output voltage test points

Near the phase indicators on the front panel are jacks into which can be plugged connectors for testing for input and output voltage. See front panel component locations. The voltage at these points are reduced by a factor of 100:1 from the actual voltage at the terminal blocks.

3.3.7 Component locations

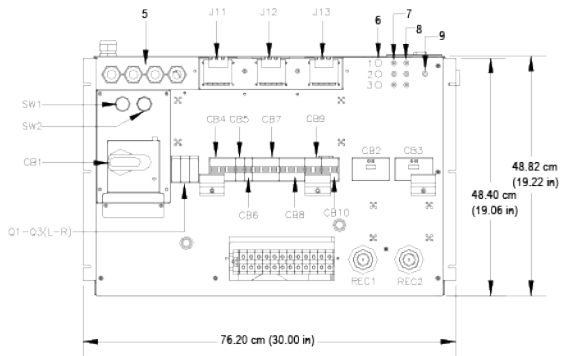
Figure 13 Top panel



1	Ground
2	Gradient 420V power

3	Spare
4	System cabinet

Figure 14 Front panel



5	Main input
6	Indicator
7	Input LEDs
8	Output LEDs
9	Ground

Chapter 4 Field Replaceable Units

4.1 Field Replaceable Units (FRUs)

No maintenance is required other than periodically (such as every 6 months) all electrical connections should be checked for tightness. Accumulations of dust or debris should be removed from the front panel to keep ventilation working for convection cooling.

IMPORTANT

All FRUs are to be replaced by authorized service personnel only.

To replace FRU components – First shut off and lock out power, remove necessary panels, locate the proper components, and replace with the new components (refer to [Table 3 on page 19](#)). Reinstall all panels and restore power.

The exact type and rating fuse must be used as the replacement, or equipment damage may occur.

Table 3 FRU listing

Item	Ref Desig	Description	GE PN	TEAL PN
1	A1	XGD PDU Control Board Assembly	5140621-2	4990780
2	Q1-Q3	TVSS Module FRU for ACGD PDU	2301200-3	4990387
3	CON1-CON2	3-Pole Power Contactor for E-stop (ABB PN A50-30-11-84, TEAL PN 0410203)	5140621-11	4990781
4	PS1-PS2	Delta 48V 2500W Power Supply	5745311	-
5	-	48V Power Supplies Bracket	5343114-10	4991419
6	FAN1-FAN2	Fan with Tach Output Sanyo PN 109E1348G101 (TEAL PN 0860028)	5140621-5	4990783
7	CB1	ABB Main Input Circuit Breaker	5604719	4991423-XGD
8	CB2	Gradient 420V Power Circuit Breaker (Eaton)	5789505	4991424

FRU listing continued				
Item	Ref Desig	Description	GE PN	TEAL PN
9	CB7, CB8	Supercon Shim Power Supply and Magnet Power Supply Circuit Breakers	5789506	4991426
		NOTE The part kit will contain all CB3, CB4, CB7, and CB8 breakers. But only CB7 and CB8 are compatible for XGD PDU. FRUs XGD CB3 and CB4 are not compatible with XGD PDU (5140621). Do not use CB3 and CB4 for replacement in XGD PDU.		
10	CB5, CB9	Brainwave and Gradient 48V Circuit Breakers	5789507	4991427
		NOTE The part kit will contain all CB5, CB6, CB9, CB10, and CB11 breakers. But only CB5 and CB9 are compatible for XGD PDU. FRUs XGD CB6, CB10, and CB11 are not compatible with XGD PDU (5140621). Do not use CB6, CB10, or CB11 for replacement in XGD PDU.		
11	REC1-2, TB2	Receptacles and Terminal Blocks	5789508	4991437
		NOTE The part kit will contain all REC1-REC3, TB1-TB3, and TB6. But only REC1-2 and TB2 are compatible for XGD PDU. FRUs XGD REC3, TB1, TB3 and TB6 are not compatible with XGD PDU (5140621). Do not use REC3, TB1, TB3, or TB6 for replacement in XGD PDU.		

FRU listing continued				
Item	Ref Desig	Description	GE PN	TEAL PN
12	SW2, TP1-7	Switches and Test Points	5789509	4991438
		NOTE The part kit will contain all IND1-IND3, SW1, SW2, and TP1-7. But only SW2 and TP1-7 are compatible for XGD PDU. FRUs IND1-3 and SW1 are not compatible with XGD PDU (5140621). Do not use IND1-3 or SW1 for replacement in XGD PDU. We do not have FRUs for XGD IND1-3 or SW1.		

4.2 FRU installation

Before moving forward with FRU replacement, you will need to identify the unit part number (PN) and revision of the unit being worked on as some FRU instructions are dependent on the unit requiring replacements. Proceed to the appropriate section that corresponds with the unit if a FRU has multiple instructions.

Figure 15 Location of unit PN and revision



Versions of the PDU changed by incrementing the part number (such as 6550012-05 to 6550012-06 to 6550012-07, and so on) and by revision. Older PDUs used alphanumeric revision numbers (A0-A9, B0-B9, and so on), while revisions made during part of 2016 and newer use numeric revision numbers (00, 01, 02, and so on). In some cases, part numbers may contain both alphanumeric and numeric revision numbers (such as A0, A1, A2, 03, 04, and so on).

4.2.1 Replacement of the TVSS module

The operation of the TVSS indicators is explained in [TVSS Indicators](#). If one (or more) of the indicators shows that a TVSS module has failed, follow the steps below to replace the failed module and restore TVSS protection:

1. Turn off the PDU circuit breaker, and follow LOTO procedures in [3.1.1 Applying LOTO - MDP on page 7](#).
2. Using a voltmeter or other voltage indicating device, check the voltage at the main input terminal board to make sure that no voltage is being applied to the input of the PDU.
3. Pull out the failed TVSS module(s) and plug in the replacement(s) as shown below.

Figure 16 TVSS module replacement



4. Restart the PDU and follow LOTO procedures in [3.1.2 Removing LOTO - MDP on page 11](#).
5. Make sure that all TVSS modules are good (no red flags displayed).

4.2.2 Removal of front panel to access additional FRUs

1. Turn off the main input circuit breaker by rotating the lever to the off position. Do [3.1.1 Applying LOTO - MDP on page 7](#).

Figure 17 Main input circuit breaker



2. Remove four (6-32 x 3/8) screws on the circuit breaker guard.

NOTE

Do not discard the screws, as they will be used to reattach the guard back over the circuit breaker.

Figure 18 Circuit breaker guard screws

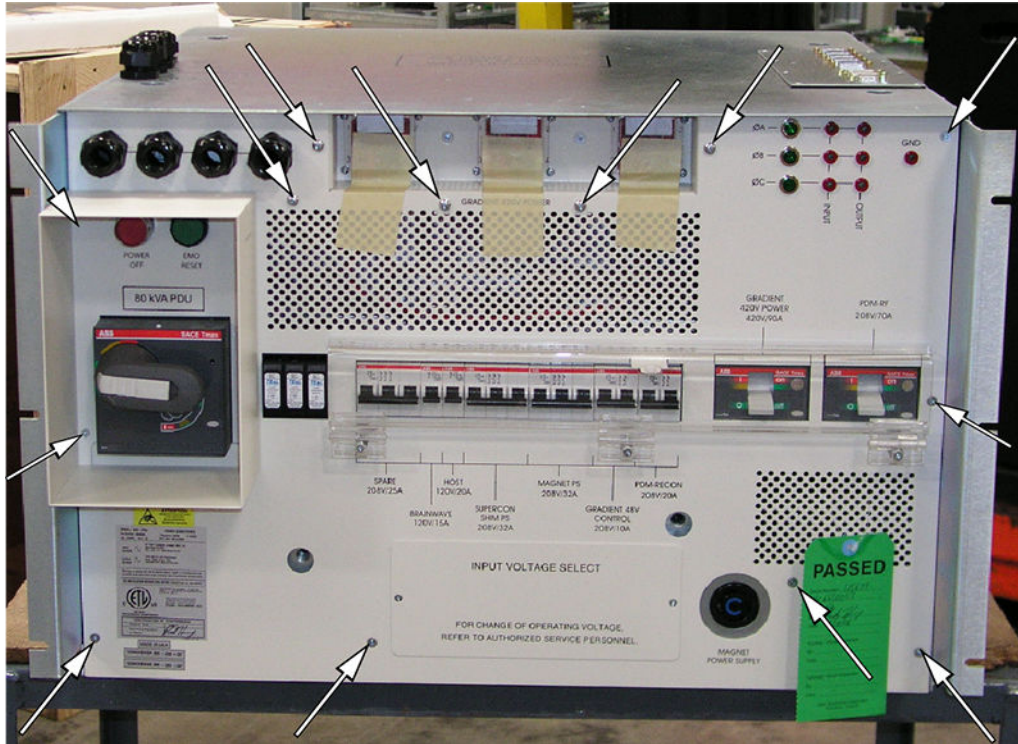


- Remove 13 (10-32 x 1/2) screws on the front panel using a Phillips screwdriver.

NOTE

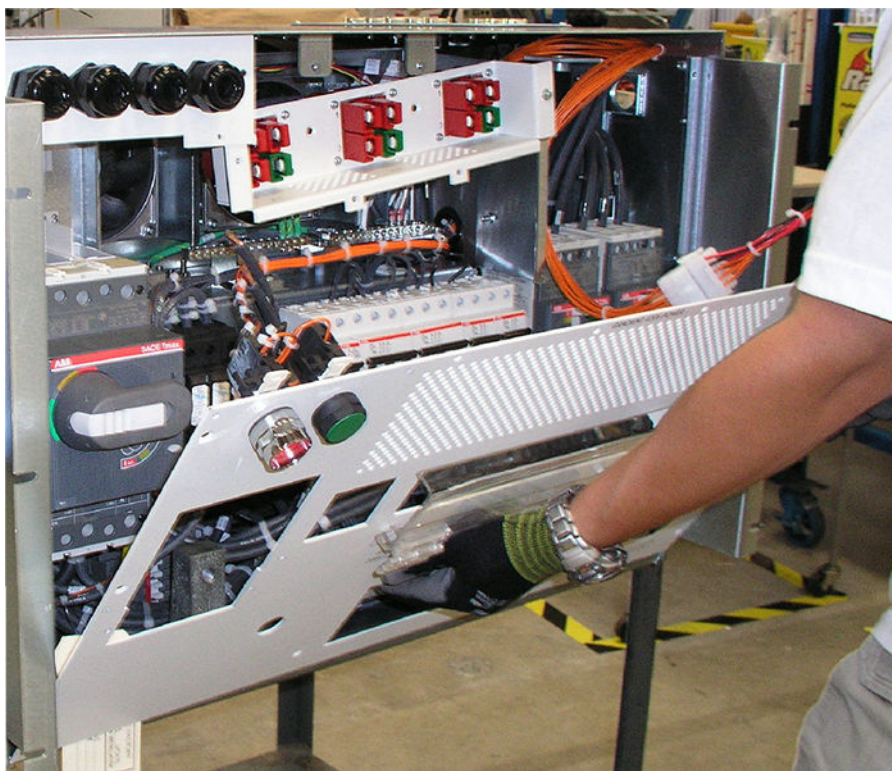
Do not discard the screws, as they will be used to reattach the front plate.

Figure 19 Front panel screws



4. Slightly pull the front panel and lean it toward the front.

Figure 20 Removing the front panel

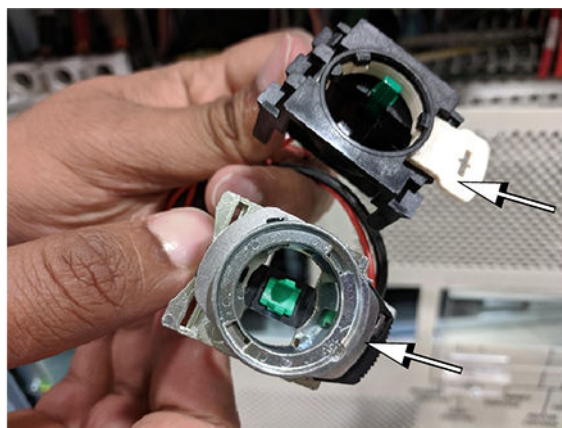
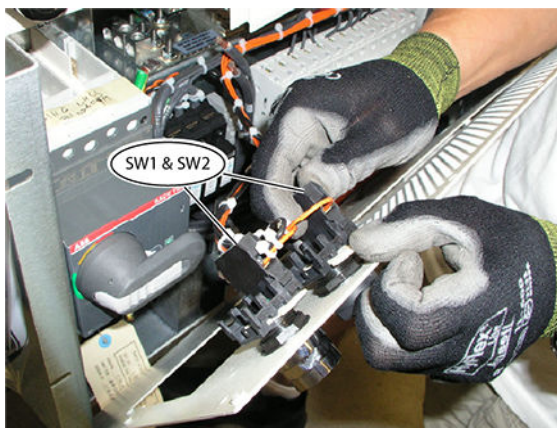


5. Turn the tabs counterclockwise and gently pull the assembly apart to disconnect SW1 and SW2 from the back of the front panel.

NOTE

No tools are required.

Figure 21 Disconnecting SW1 and SW2 (left) and SW tabs (right)



6. Press the locking tab and pull the connectors apart to disconnect P8 and J8 from the 15-pin housing connector.

NOTE

No tools are required.

Figure 22 15-pin housing connector (P8/J8)



4.2.3 Replacement of the control PC board

If the control board is deemed defective per the [4.2.14 Troubleshooting guide on page 56](#), follow the steps below to replace the control board. The control board is mounted on the inside of the side panel.

Circuit boards or assemblies with static sensitive devices will be packaged in accordance to prevent damage due to electrostatic discharge (ESD). Procedures to prevent ESD should be followed when handling static sensitive devices.

1. Remove the two mounting screws (8-32 x 3") from the circuit breaker (CB2).

NOTE

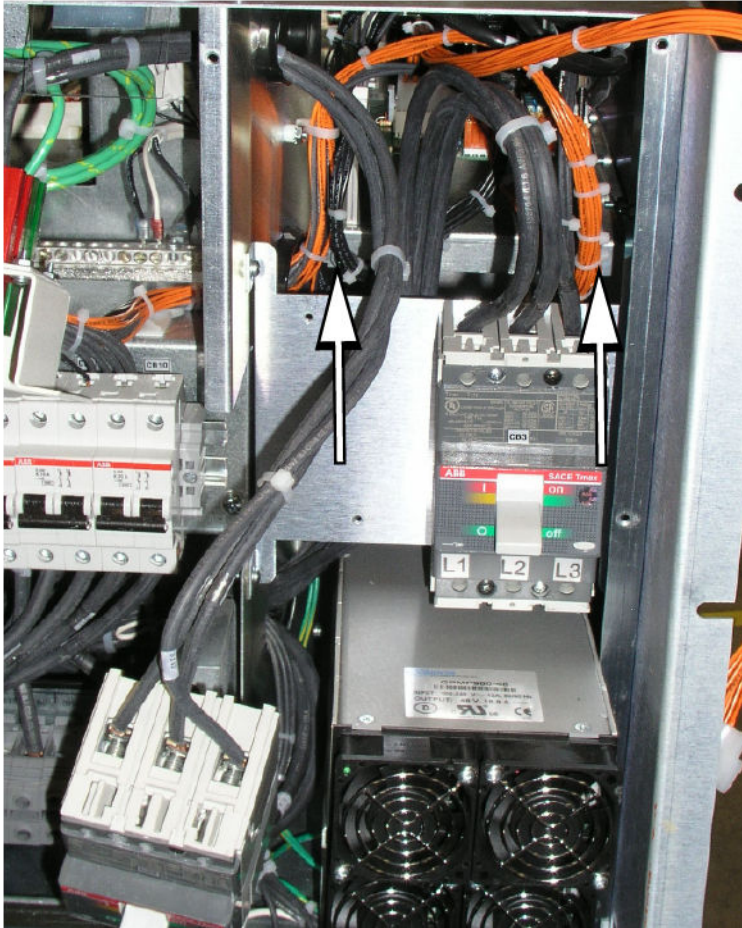
Do not discard screws, as they will be used to replace CB.

Figure 23 Mounting screws in the circuit breaker



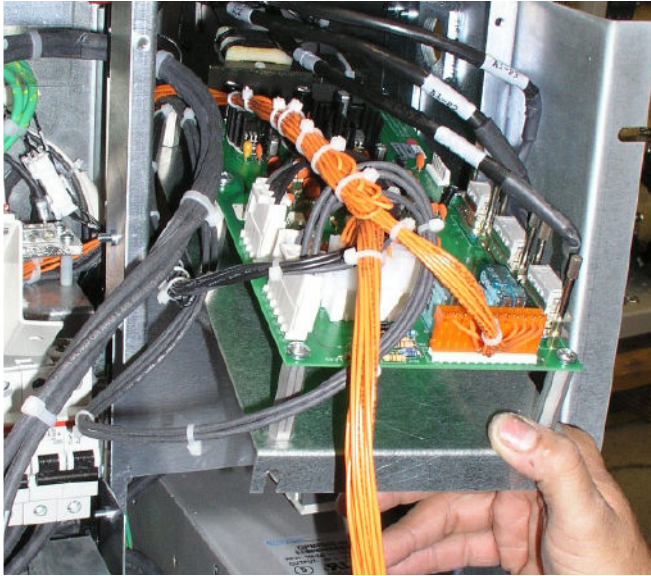
2. Loosen (do not remove) the two screws (10-32 x 1/2) at the PCB.

Figure 24 PCB screws



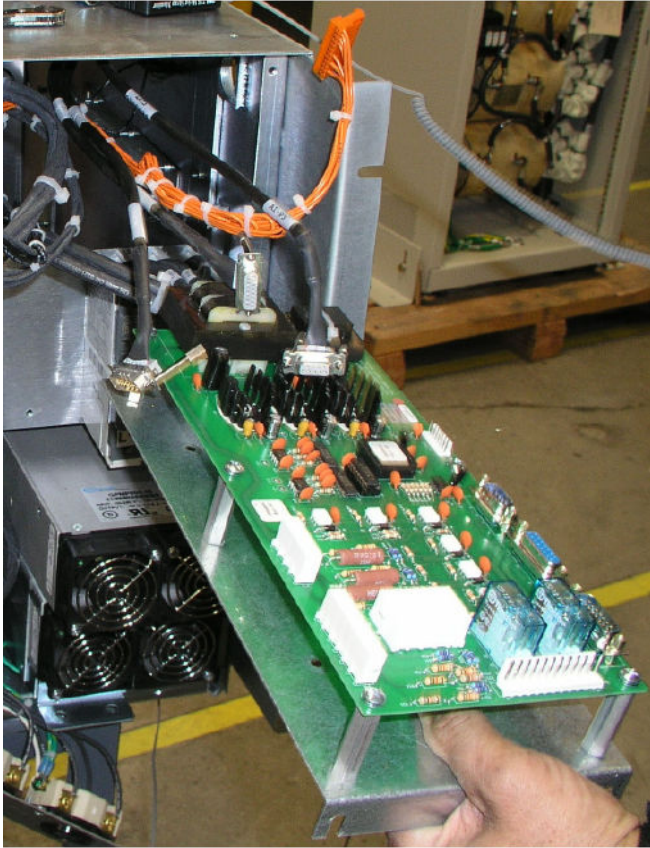
3. Carefully lift up the PCB assembly and pull it out halfway.

Figure 25 Lifting the PCB assembly



4. Disconnect cables P1, P2, P3, A1-P5, A1-P6, A1-P7, and A1-P8 from the PCB assembly. Mark the cables with masking tape and a marker to make sure of the correct position when reconnecting the cables.

Figure 26 Cables are disconnected from the PCB assembly



Reference the following figures for the placement of each cable.

Figure 27 Cable placement

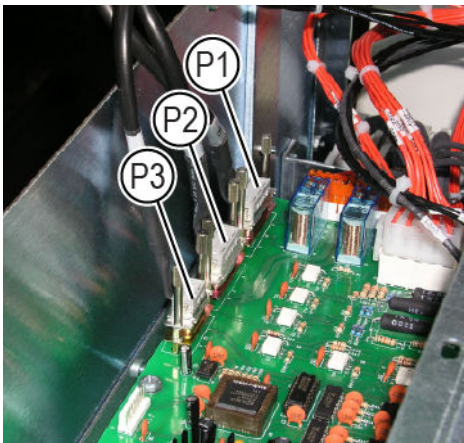
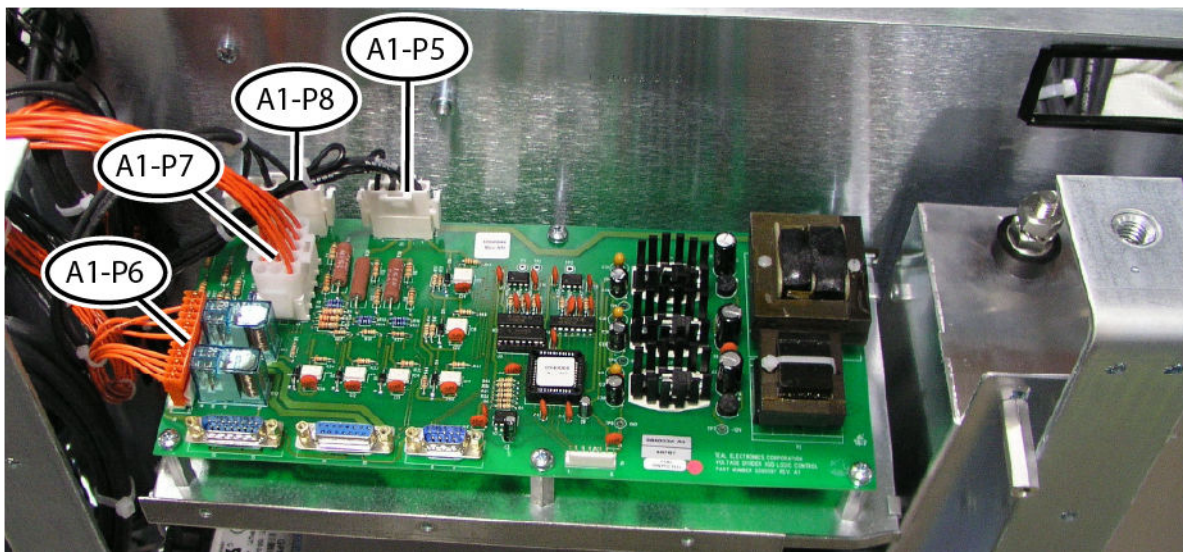
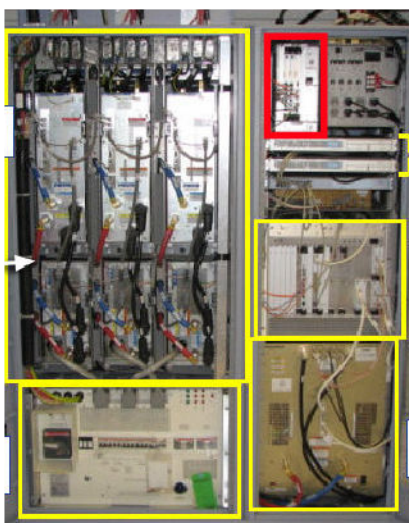


Figure 28 Cable placement

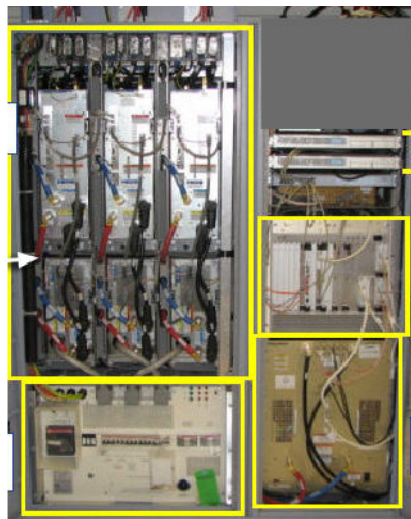
5. Configure the replacement of the PDU control board according to the cabinet type. All boards should be shipped with the board set to the AUX position. Do [Step 1 on page 26](#) to [Step 4 on page 30](#) in reverse order to replace the board.

XGD and XGDII configuration settings

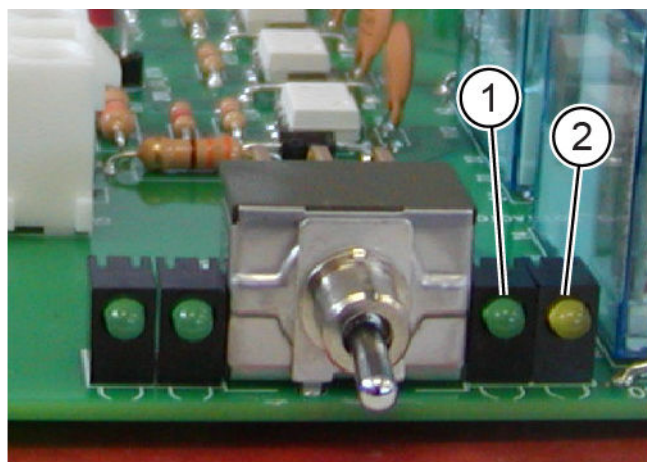
Figure 29 XGD cabinet with GMC (in red)

Board part number	Jumper / switch	LED color	Position
5343114-2	JP1 and JP2	N/A	1-2

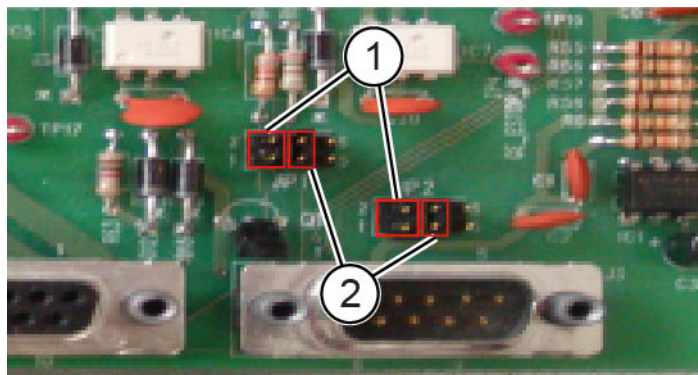
Board part number	Jumper / switch	LED color	Position
5343114-5	SW1	Green	Switch is up when viewed from the front.

Figure 30 XGD II Cabinet (without GMC)

Board part number	Jumper / Switch	LED Color	Position
5343114-2	JP1 and JP2	N/A	3-4
5343114-5	SW1	Yellow	Switch is down when viewed from the front.

Figure 31 5343114-5 control board

1	(For XGD configuration) Switch is up. LED 3 is green.
2	(For XGD II configuration) Switch is down. LED 4 is yellow.

Figure 32 5343114-2 control board

1	(For XGD configuration) Jumper 1 and 2
2	(For XGD II configuration) Jumper 3 and 4

1. Do [3.1.2 Removing LOTO - MDP on page 11](#).
2. Once powered back up, inspect the control board and make sure the appropriate configuration LED is on. This LED is located above the configuration switch. Green indicates XGD configuration and yellow indicates XGD II configuration.
3. Upon replacement, make sure the E-stop circuitry is functional using the GE Emergency Stop and Off procedure.

NOTE

If a control board is placed in the AUX position and installed in an XGD II cabinet, a disconnected PEM – PDU cable will not fail safe and issue an Emergency Off condition. This means the main PDU breaker will not shunt trip if there is a leak in the cabinet. If a control board is placed in the PEM position and installed in an XGD cabinet, the PDU control board will issue an Emergency Off once it powers up. This means the field engineer will be unable to power up the PDU.

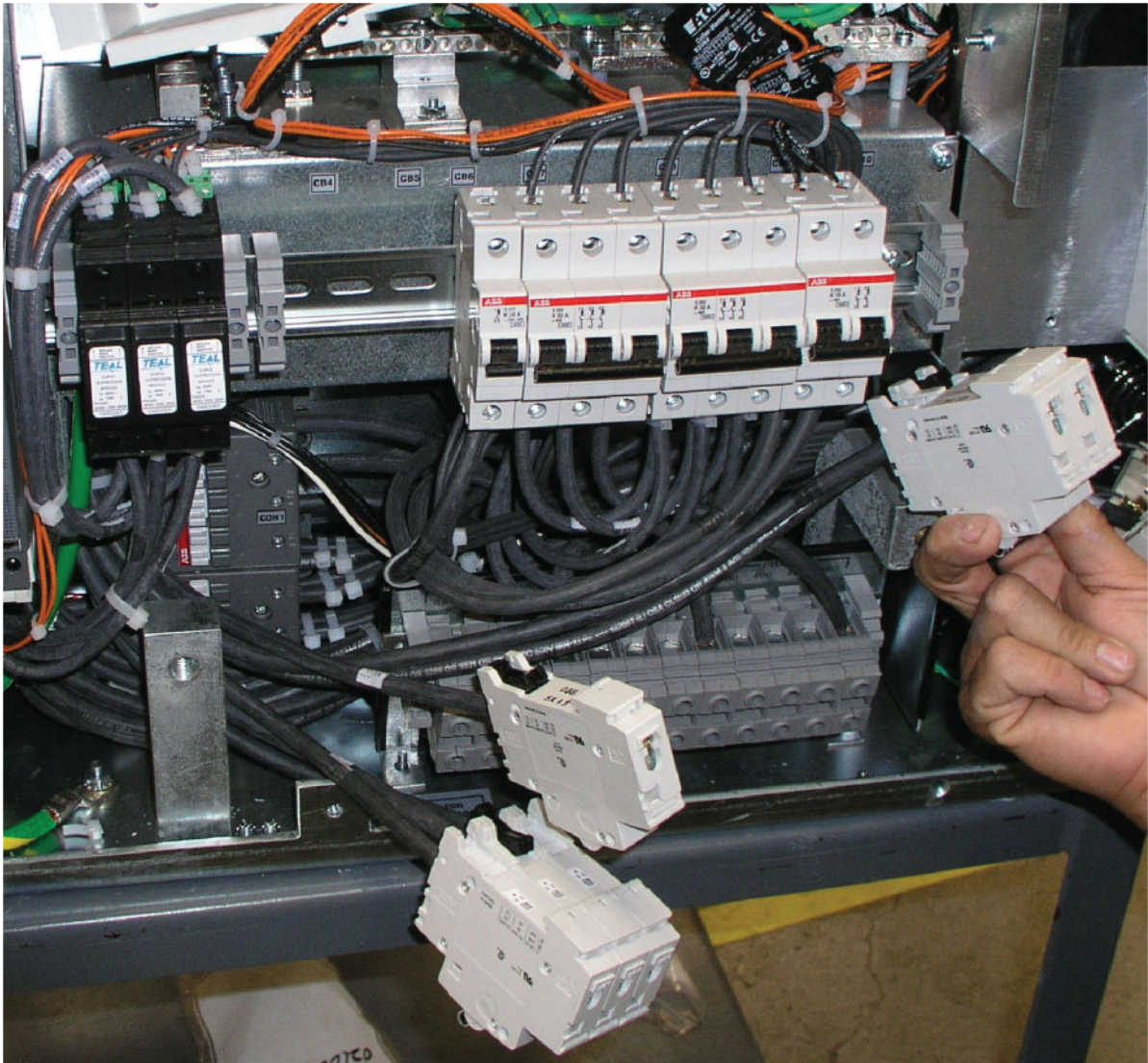
4.2.4 Replacement of contactors (CON1 and CON2)

If the contactors are deemed defective per the [Troubleshooting Guide](#), follow the steps below to replace the contactors. The contactors are mounted behind the circuit breaker panel of the PDU located on the front panel.

Contactors or assemblies with static sensitive devices will be packaged in accordance to prevent damage due to electrostatic discharge (ESD). Procedures to prevent ESD should be followed when handling static sensitive devices.

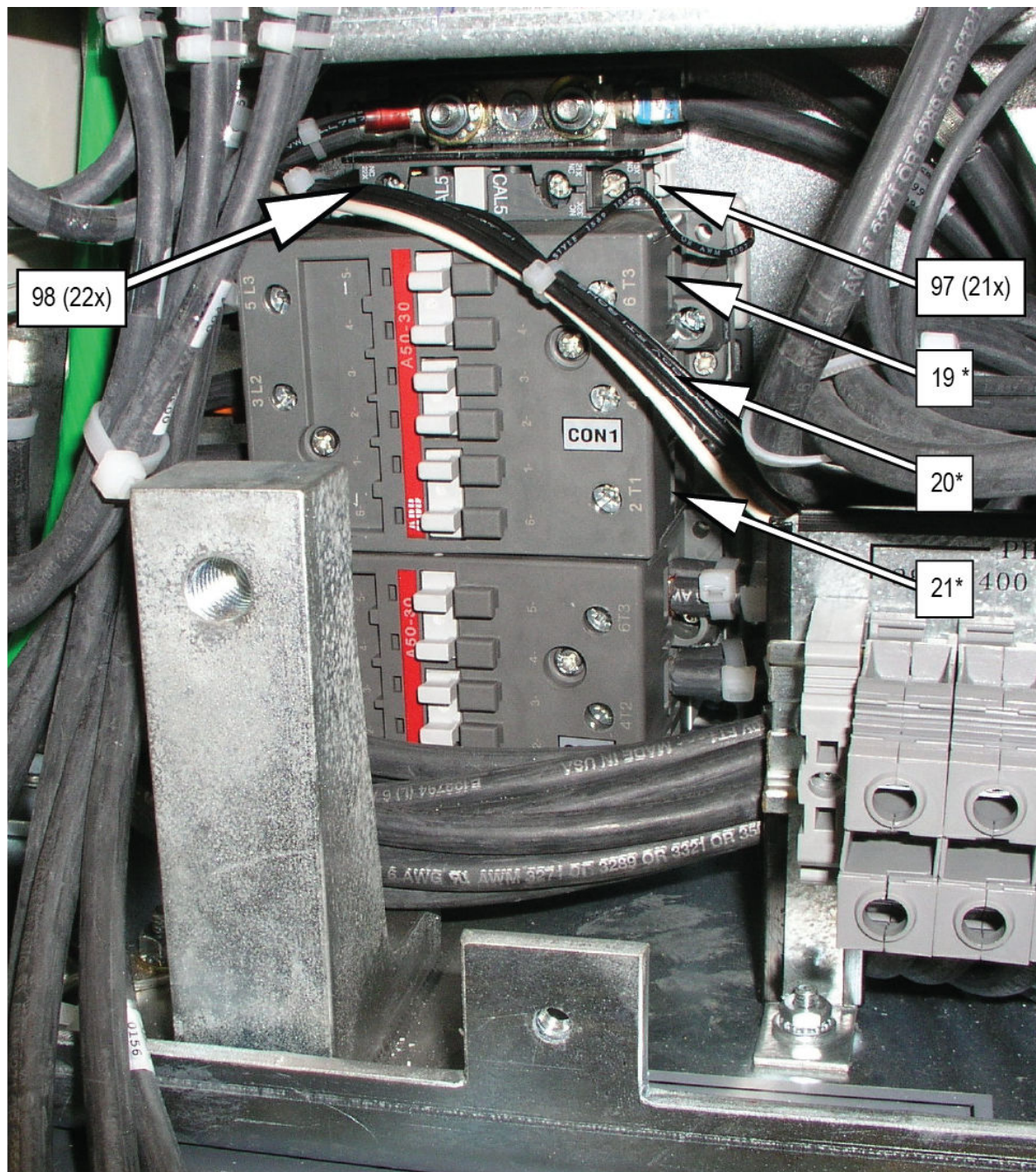
1. Remove (snap off) circuit breakers CB4, CB5, and CB10 from the mounting bracket.

Figure 33 Circuit breakers CB4, CB5, and CB10



2. Disconnect wires 97 and 98 from the AUX of CON1 and wires 19 to 21 from T1 to T3 of CON1. Make sure you mark the wires with masking tape and a marker.

Figure 34 CON1 wire connections



*Wires 19, 20, and 21 are not shown.

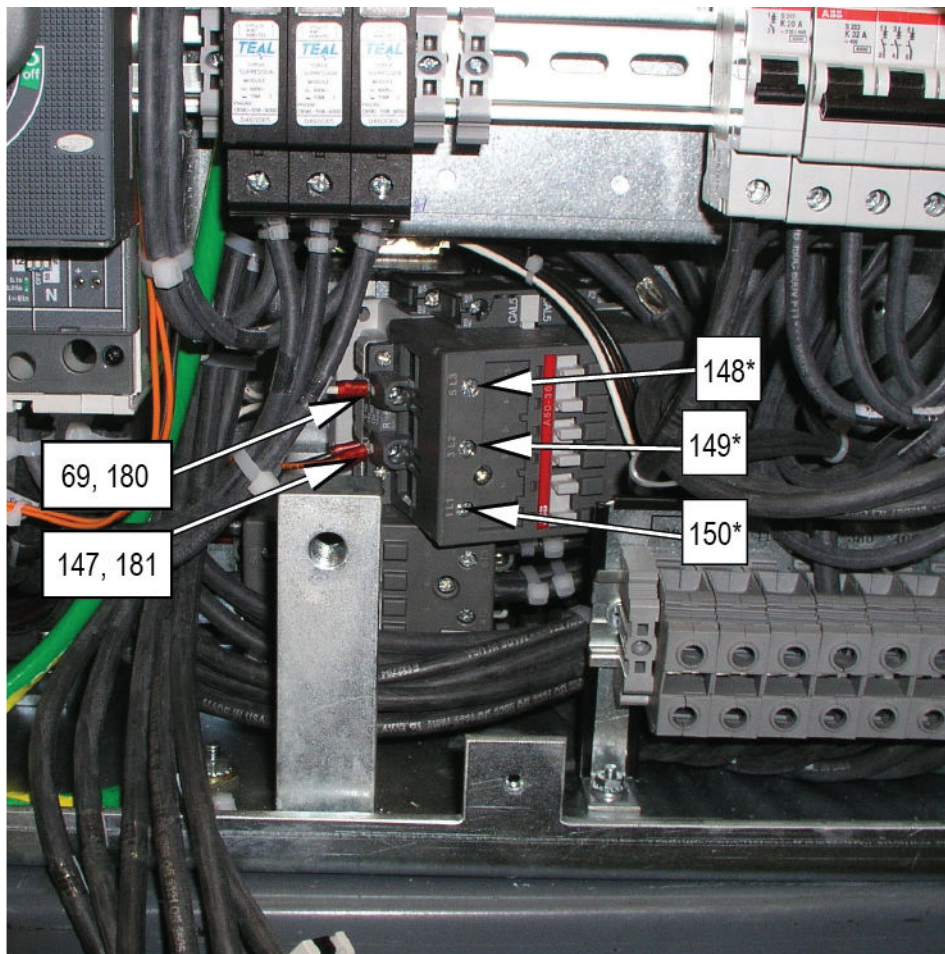
3. Snap out CON1 from the mounting bracket using the tab at the base of the contactor and disconnect wire 148 to 150 from L1 to L3.

Figure 35 Pull the tab to release CON1 from the din rail



4. Disconnect wires 69 and 180 from A1 of CON1 and wires 147 and 181 from A2 of CON1. Make sure you mark the wires with masking tape and a marker.

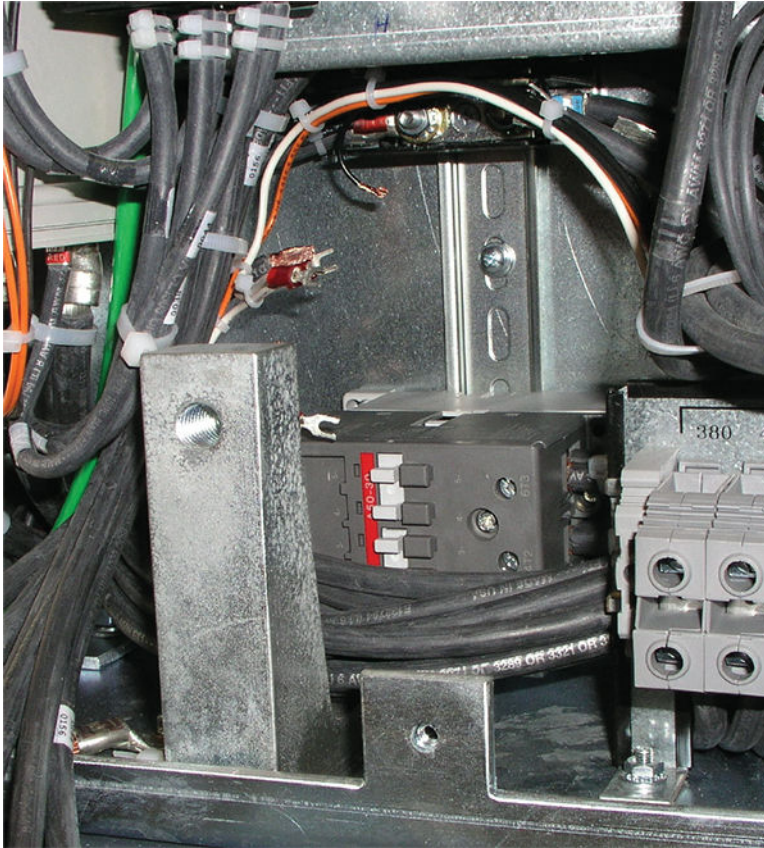
Figure 36 CON1 wire connections



* Wires 148, 149, and 150 are not shown.

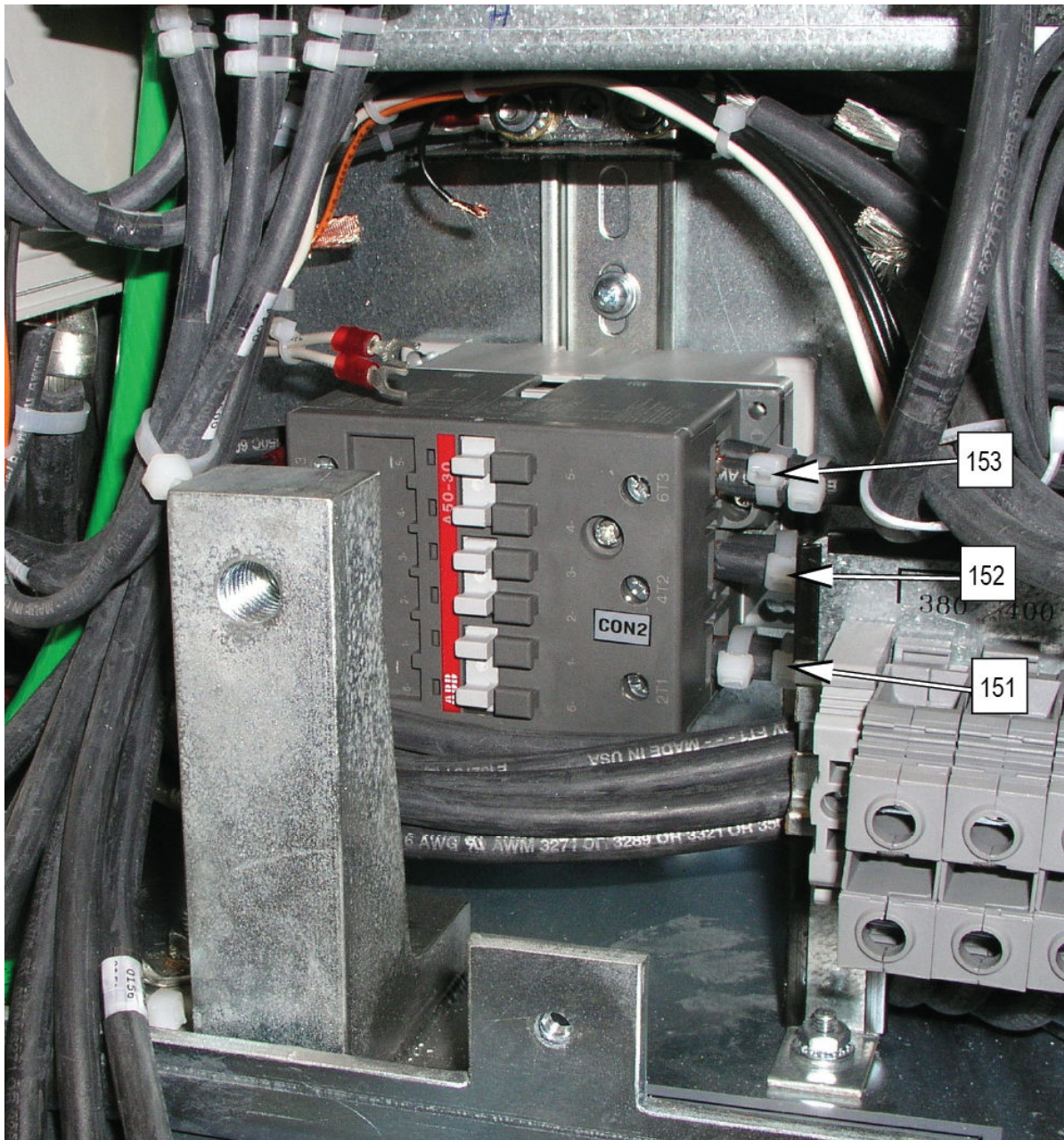
5. Remove CON1 from unit and proceed to removing CON2.

Figure 37 CON1 removed from unit



6. Disconnect wire 151 from L1, 152 from L2 and 153 from L3 of CON2.

Figure 38 CON2 wire connections



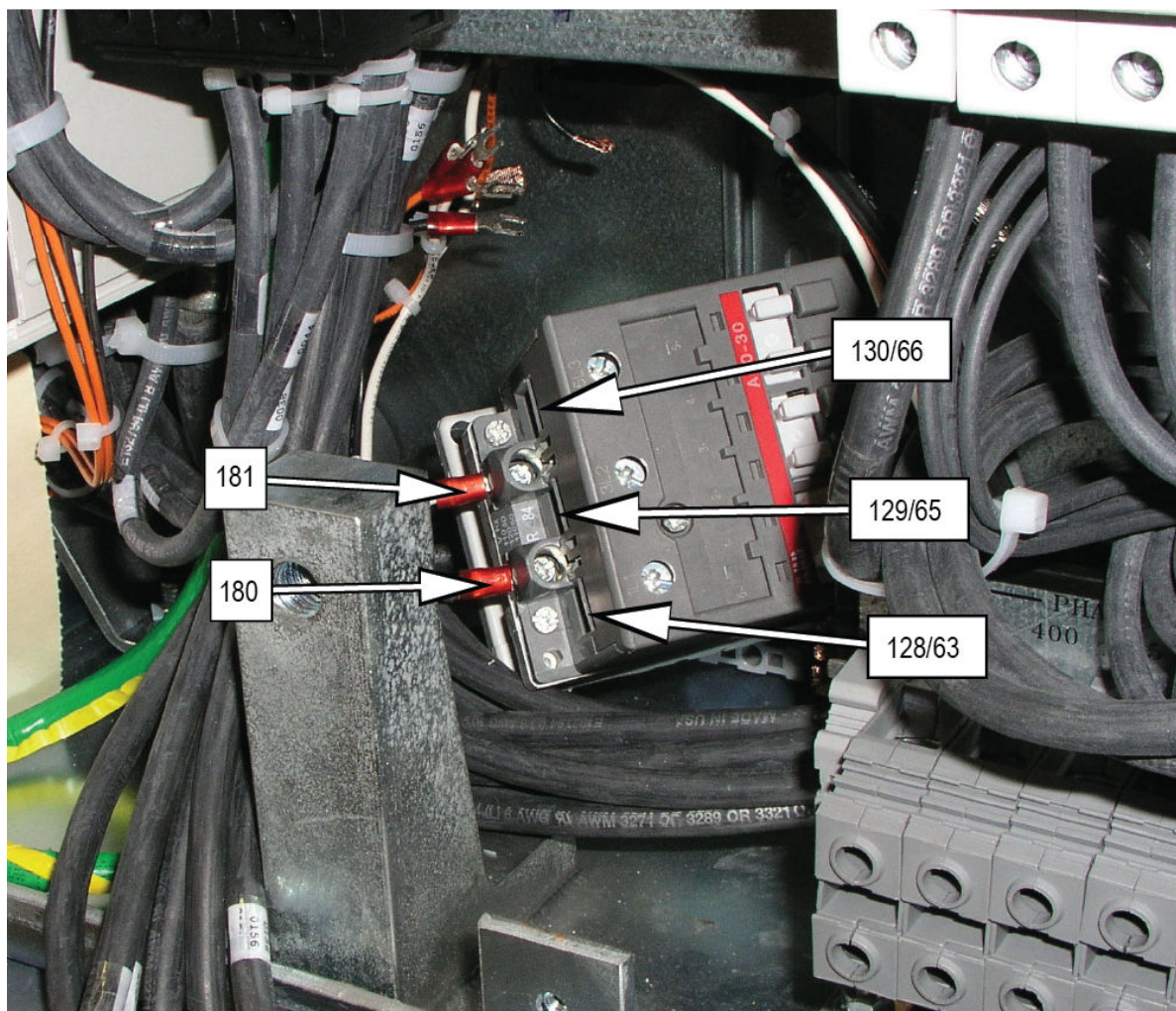
7. Snap out CON2 using the tab at the base of the contactor, and disconnect wires 128 and 63 from T1, 65 and 129 from T2, and 130 and 66 from T3 of CON2.

Figure 39 Pull the tab to release CON2 from the din rail



8. Disconnect wire 180 from A1 and 181 from A2 of CON2.

Figure 40 CON2 wire connections



9. Remove CON2 from the unit and prepare to install the replacement parts.
10. Replace old contactor with a new contactor.
 - a. Reconnect the wires on CON1 and/or CON2 based on the schematic diagram.
 - b. Tighten all connections per the torque specifications below.

Table 4 Torque specification

Connections	Pound per inch
T1-T3	35
L1-L3	35

Torque specification continued	
Connections	Pound per inch
AUX	9
A1-A2	9

4.2.5 Replacement of fans

If the fans are deemed defective per the [Troubleshooting Guide](#), follow the steps below to replace the fans. The fans are mounted behind the circuit breaker panel of the PDU located on top of the unit.

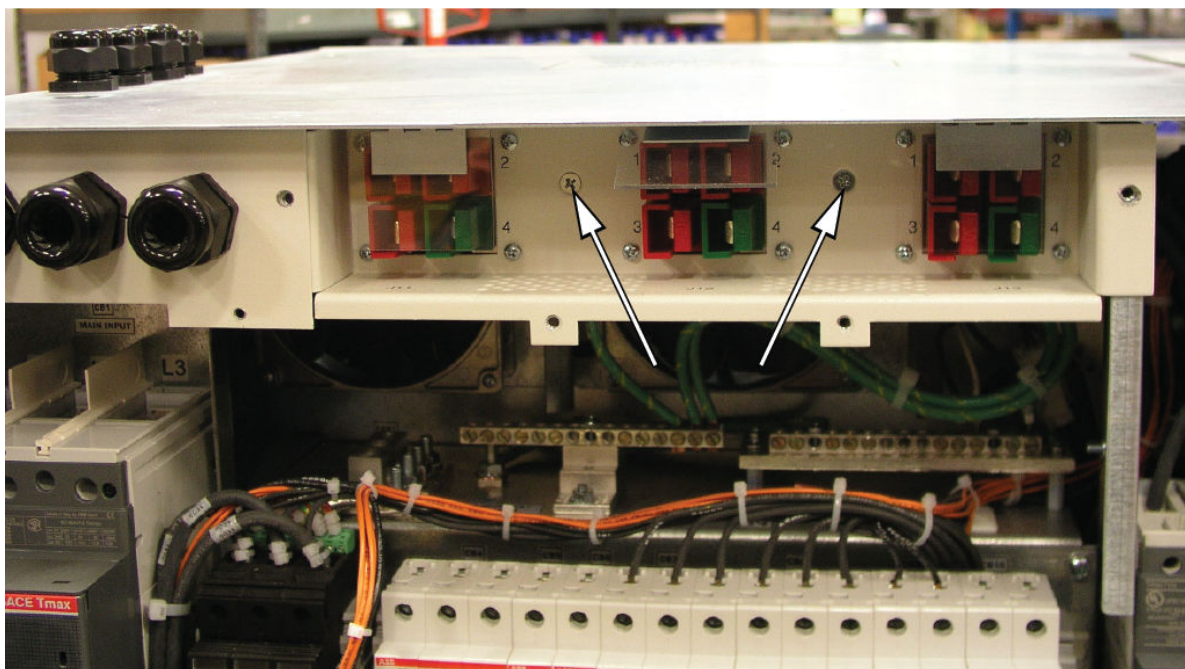
Fans or assemblies with static sensitive devices will be packaged in accordance to prevent damage due to electrostatic discharge (ESD). Procedures to prevent ESD should be followed when handling static sensitive devices.

1. Remove two (10-32 x 1/2) screws from the Anderson assembly connector.

NOTE

Do not discard the screws, as they will be used to replace the assembly.

Figure 41 Anderson assembly connector screws

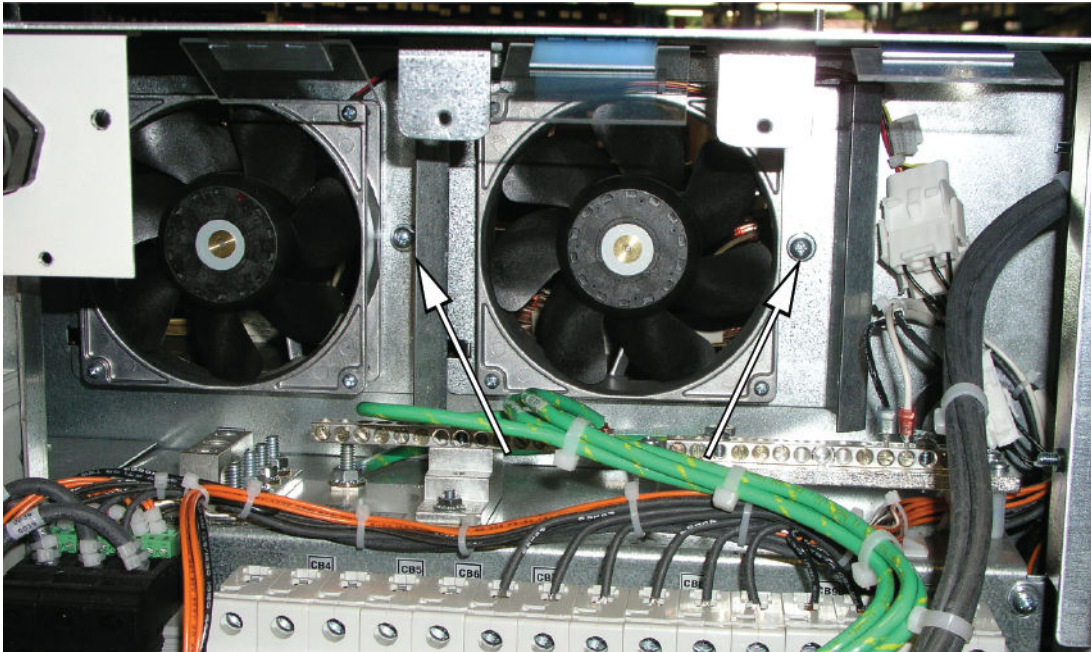


2. Remove the fans (FAN1 and FAN2) by unscrewing two (10-32 x 1/2) screws using a Phillips screwdriver.

NOTE

Do not discard the screws, as they will be used to replace the fan assembly.

Figure 42 Screws securing the fan assembly



3. Unplug the fan connectors and remove the fan assembly from the unit.

Figure 43 Unplugging the fan connectors

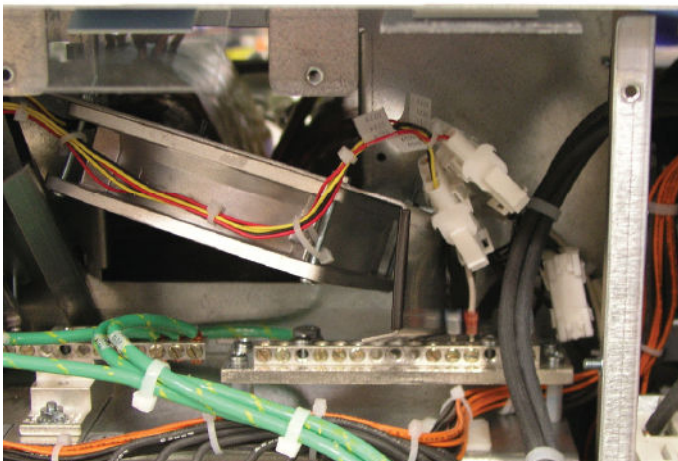
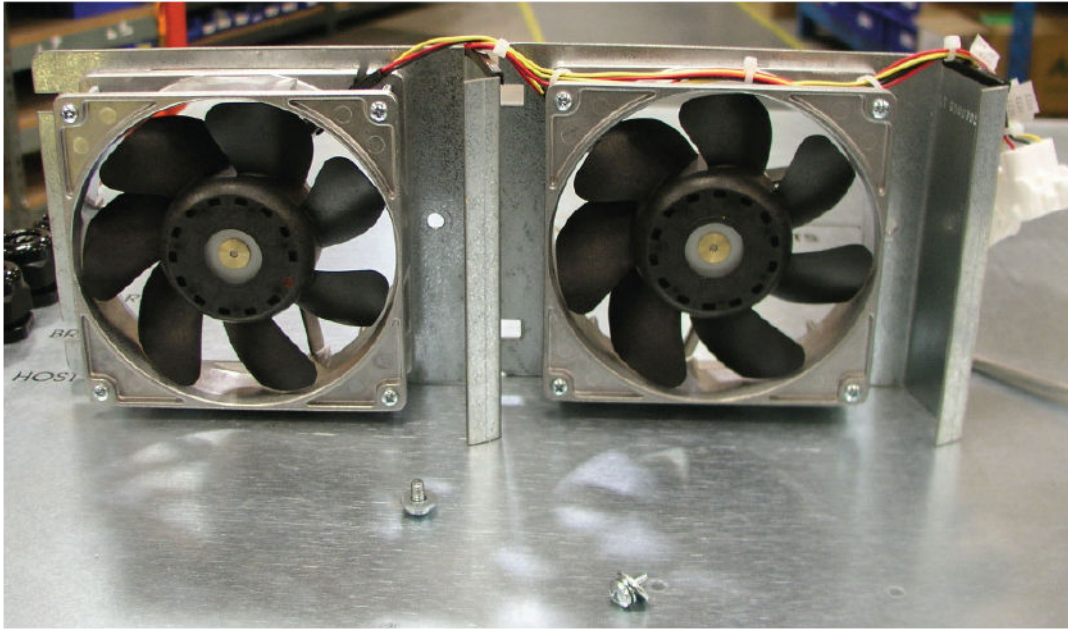


Figure 44 Fan assembly



4. Replace the old fan assembly with the FRU using the screws previously removed.
5. Reconnect the housing connectors P14 with J17 and P19 with J19.

4.2.6 Replacement of power supplies

If the power supplies are deemed defective per the [4.2.14 Troubleshooting guide on page 56](#), follow the steps outlined in [Appendix A - Replacement of power supplies](#). The power supplies are mounted behind the receptacle assembly of the PDU located on bottom of the unit.

4.2.7 Replacement of the main input circuit breaker (CB1)

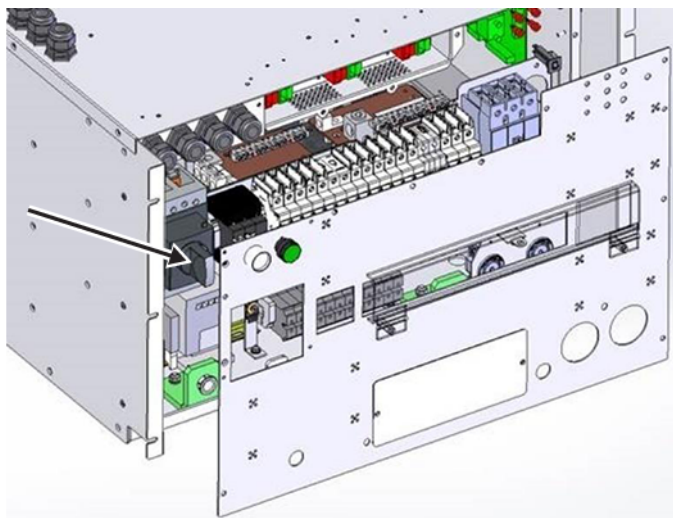
Before moving forward with circuit breaker replacement, you will need to identify the unit part number (PN) and revision of the unit being worked on. Proceed to the appropriate section below that corresponds with the unit.

Figure 45 Unit part number and revision

If the circuit breaker is deemed defective per the [4.2.14 Troubleshooting guide on page 56](#), follow the steps below to replace the circuit breaker.

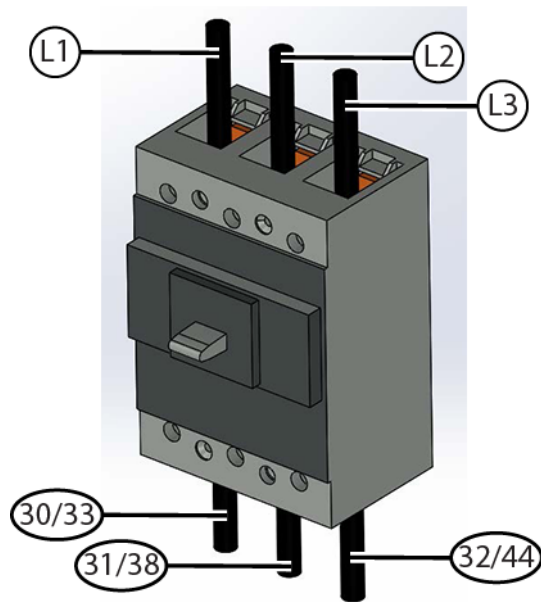
4.2.7.1 Replacing CB1 for units 6550012-07 Rev A1 and prior [manufactured before 5 October 2017 (ABB breaker)]

1. Remove the front panel per [4.2.2 Removal of front panel to access additional FRUs on page 23](#).

Figure 46 CB1 location under the front panel

2. Disconnect wires 30/33, 31/38, and 32/44 from the output (bottom) of the ABB circuit breaker CB1 using a hex key; disconnect wires L1, L2, and L3 from the input (top) of the breaker. Label the wires with masking tape if necessary.

Figure 47 CB1 wire connections

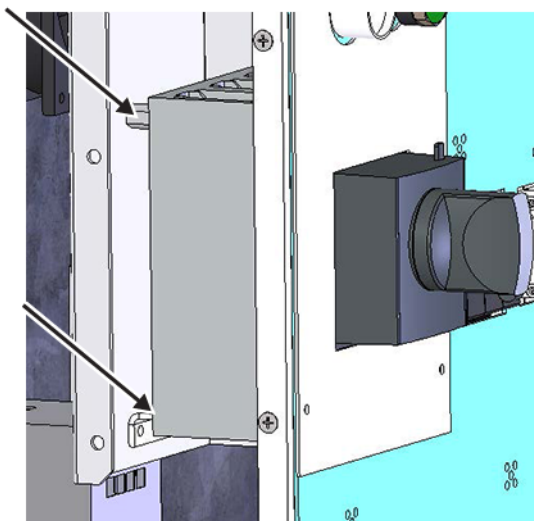


3. Cut shunt trip wires 70 and 73.
4. Remove the four mounting screws holding the circuit breaker to the plate.

NOTE

Do not discard the screws, as they will be used to attach the new circuit breaker.

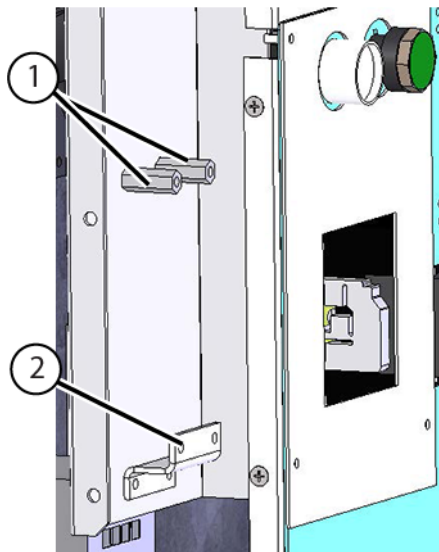
Figure 48 CB1 mounting screws*



There are two screws in each location.

5. Remove the circuit breaker from the mounting plate.
6. Install the provided standoffs and bracket to hold the new circuit breaker in place.

Figure 49 Standoffs and bracket



1	Standoffs
2	Bracket

7. Install the new ABB circuit breaker in place and secure it with the four mounting screws.
8. Butt splice the new shunt trip wires.
9. Reconnect the wires that were disconnected in [Step 2 on page 46](#).

4.2.8 Replacement of the gradient 420V power circuit breaker (CB2)

Before moving forward with circuit breaker replacement, you will need to identify the unit part number (PN) and revision of the unit being worked on.

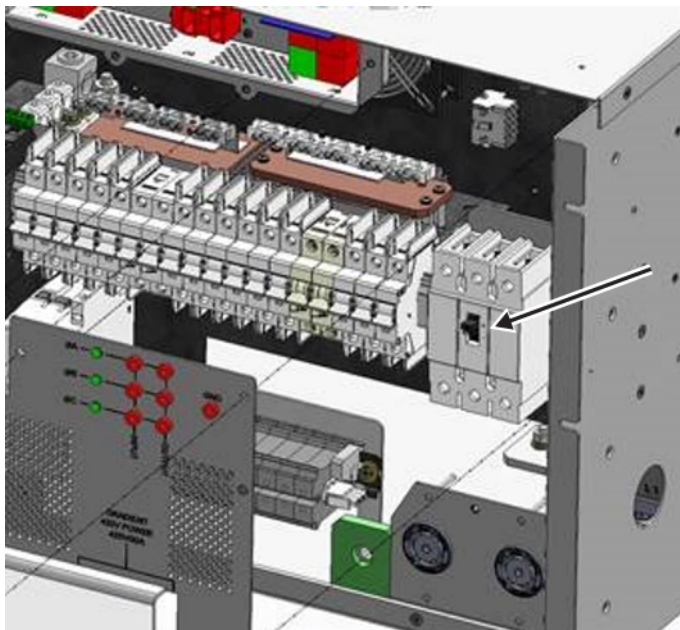
Figure 50 Unit part number and revision



If the circuit breaker is deemed defective per the [4.2.14 Troubleshooting guide on page 56](#), follow the steps below to replace the circuit breaker.

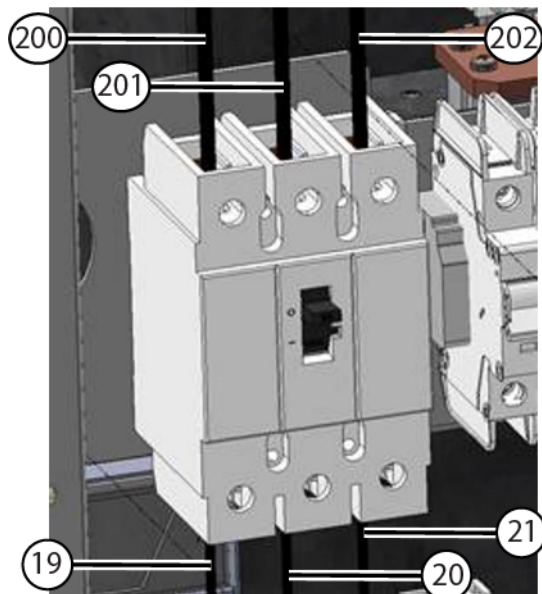
1. Remove the front panel per [4.2.2 Removal of front panel to access additional FRUs on page 23](#).

Figure 51 CB2 location under the front panel



2. Using a flathead screwdriver, disconnect wires 19, 20, and 21 from the input and 200, 201, and 202 from the output of the Eaton circuit breaker CB2. Label the wires with masking tape if necessary.

Figure 52 CB2 wire connections



3. Using a flathead screwdriver, remove the four mounting screws holding the circuit breaker to the plate.

NOTE

Do not discard the screws, as they will be used to attach the new circuit breaker.

4. Remove the circuit breaker from the mounting plate.
5. Install the new Eaton circuit breaker in place and secure it with the four mounting screws.
6. Reconnect the wires that were disconnected in [Step 2 on page 49](#).

4.2.9 Replacement of superconducting shim power supply and magnet power supply circuit breakers (CB7 and CB8)

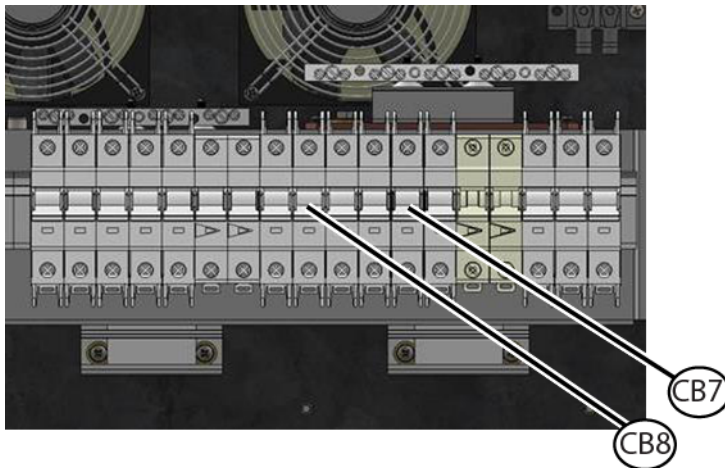
NOTE

The part kit will contain all CB3, CB4, CB7, and CB8 breakers. But only CB7 and CB8 are compatible for XGD PDU.

FRUs XGD CB3 and CB4 are not compatible with XGD PDU (5140621). Do not use CB3 and CB4 for replacement in XGD PDU.

1. Remove the front panel per [4.2.2 Removal of front panel to access additional FRUs on page 23](#).
2. Disconnect wires 134, 135, and 136 from the input (bottom) of the Eaton circuit breaker CB7; disconnect wires 140, 141, and 142 from the output (top) of the circuit breaker. Label the wires with masking tape if necessary.

Figure 53 Circuit breaker locations



3. Remove the circuit breaker from the DIN rail by gently pulling on the tab on the bottom of the circuit breaker using a flathead screwdriver.
4. Install the new Eaton circuit breaker on the DIN rail.
5. Disconnect wires 131/134, 132/135, and 133/136 from the input (bottom) of the Eaton circuit breaker CB8; disconnect wires 137, 138, and 139 from the output (top) of the circuit breaker. Label the wires with masking tape if necessary.
6. Repeat [Step 3 on page 50](#) and [Step 4 on page 50](#) for CB8.
7. Reconnect the wires that were disconnected.

4.2.10 Replacement of BrainWave and gradient 48V circuit breakers (CB5 and CB9)

NOTE

The part kit will contain all CB5, CB6, CB9, CB10, and CB11 breakers. But only CB5 and CB9 are compatible for XGD PDU.

FRUs XGD CB6, CB10, and CB11 are not compatible with XGD PDU (5140621). Do not use CB6, CB10, or CB11 for replacement in XGD PDU.

1. Remove the front panel per [4.2.2 Removal of front panel to access additional FRUs on page 23](#).
2. Disconnect wire 210 from the input (bottom) of the Eaton circuit breaker CB5; disconnect wire CB5 from the output (top) of the circuit breaker. Label the wires with masking tape if necessary.

Figure 54 Circuit breaker locations



3. Remove the circuit breaker from the DIN rail by gently pulling on the tab on the bottom of the circuit breaker using a flathead screwdriver.
4. Install the new Eaton circuit breaker on the DIN rail.
5. Disconnect wires 26 and 27 from the input (bottom) of the Eaton circuit breaker CB9; disconnect wires 89 and 90 from the output (top) of the circuit breaker. Label the wires with masking tape if necessary.
6. Repeat [Step 3 on page 51](#) and [Step 4 on page 51](#) for CB9.
7. Reconnect the wires that were disconnected.

4.2.11 Replacement of receptacles and terminal blocks (REC1-2 and TB2)

Access to some FRUs might be difficult. Additional components within the cabinet will need to be removed and/or the PDU removed from the cabinet in order to access the FRU if replacement is desired.

NOTE

The part kit will contain all REC1-REC3, TB1-TB3, and TB6. But only REC1-2 and TB2 are compatible for XGD PDU.

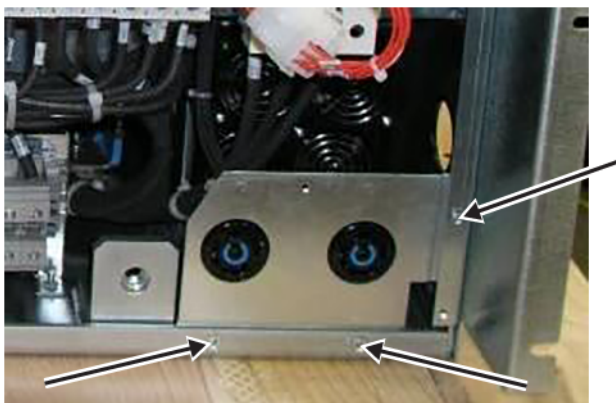
FRUs XGD REC3, TB1, TB3 and TB6 are not compatible with XGD PDU (5140621). Do not use REC3, TB1, TB3, or TB6 for replacement in XGD PDU.

1. Remove the front panel per [4.2.2 Removal of front panel to access additional FRUs on page 23](#).

4.2.11.1 Replacing the REC1 and REC2 receptacles

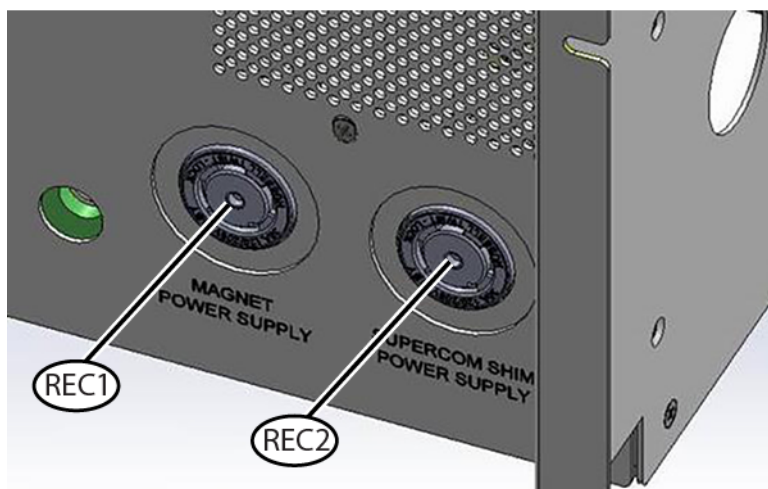
1. Remove the three (10-32 x 1/2) screws from the receptacle assembly.

Figure 55 Receptacle assembly screws



2. Disconnect wires 137 (X), 138 (Y), 139 (Z), 144 (W), and 143 (G) from REC1. Label the wires with masking tape if necessary.

Figure 56 Receptacle locations



3. Disconnect wires 140 (X), 141 (Y), 142 (Z), 144/146 (W), and 1433/145 (G) from REC2. Label the wires with masking tape if necessary.

4. Remove the screws securing REC1 and REC2, and remove the receptacles.
5. Install the new REC1 and REC2, and secure them with the screws.
6. Reconnect the wires that were disconnected.
7. Use the four (10-32 x 1/2) screws to secure the receptacle assembly.

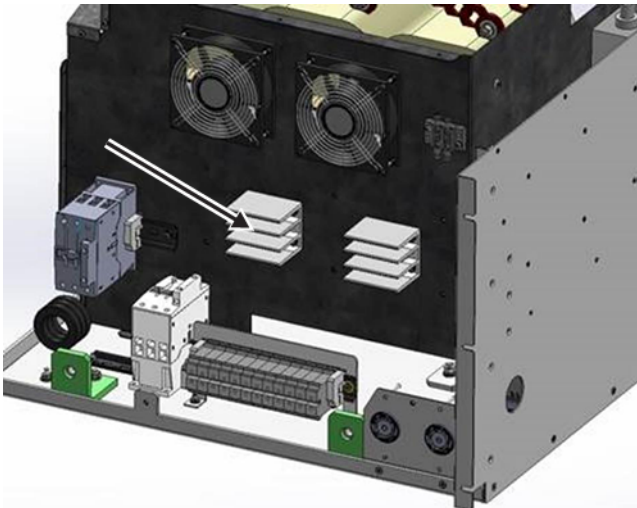
4.2.11.2 Replacing the TB2 terminal block

NOTE

Access is difficult unless the panel of mini CBs is removed.

1. Disconnect wires 24, 28/59, and 29/60 from the input (left) of TB2. Label the wires with masking tape if necessary.
2. Remove the screws securing TB2.

Figure 57 TB2 location



3. Remove TB2 from the panel.
4. Install the new TB2.
5. Reconnect the wires that were disconnected.

4.2.12 Replacement of switches and test points (SW2 and TP1-7)

NOTE

The part kit will contain all IND1-IND3, SW1, SW2, and TP1-7. But only SW2 and TP1-7 are compatible for XGD PDU.

FRUs IND1-3 and SW1 are not compatible with XGD PDU (5140621). Do not use IND1-3 or SW1 for replacement in XGD PDU. We do not have FRUs for XGD IND1-3 or SW1.

1. Remove the front panel per [4.2.2 Removal of front panel to access additional FRUs on page 23](#).

4.2.12.1 Replacing the SW2 switch

1. Disconnect wires 74 and 75 from SW2.

Figure 58 Switch location

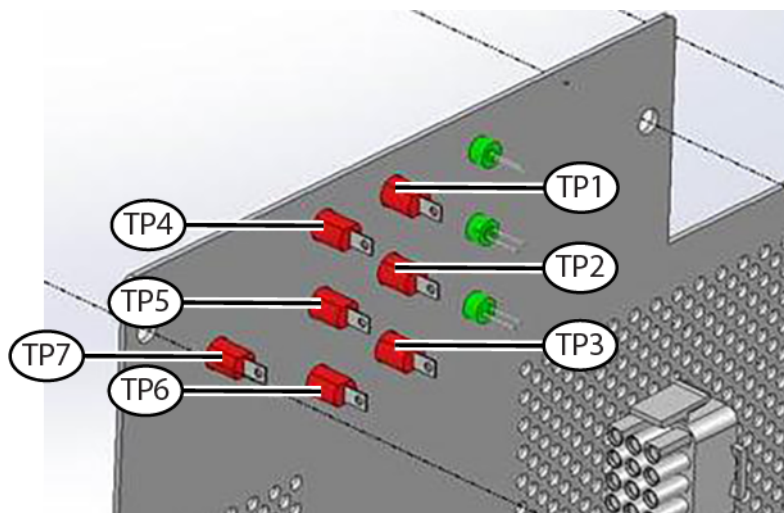


2. Disassemble SW2 to remove it from the front panel by unscrewing the nut on the back of the switch.
3. Install the new SW2 in place.
4. Reconnect the wiring removed in [Step 1 on page 54](#).

4.2.12.2 Replacing the test points

1. Disconnect wire 185 from TP1.

Figure 59 Test point locations



2. Remove the nut on the back of TP1 and slide TP1 out of the front panel.
3. Install the new TP1 and reconnect the wire removed in [Step 1 on page 55](#).
4. Repeat the previous steps for TP2 through TP7 using the table below as a reference for the correct wire numbers.

Table 5 Test point wire numbers

Test point	Wire number
TP2	186
TP3	187
TP4 (ground)	188
TP5	189
TP6	190
TP7	191

4.2.13 Placement of new FRUs

1. To replace any of the following FRUs, follow the instructions in reverse order:
 - [4.2.1 Replacement of the TVSS module on page 22](#)
 - [4.2.3 Replacement of the control PC board on page 26](#)
 - [4.2.4 Replacement of contactors \(CON1 and CON2\) on page 33](#)

- [4.2.5 Replacement of fans on page 42](#)
 - [4.2.6 Replacement of power supplies on page 44](#)
 - [4.2.7 Replacement of the main input circuit breaker \(CB1\) on page 44](#)
 - [4.2.8 Replacement of the gradient 420V power circuit breaker \(CB2\) on page 47](#)
 - [4.2.9 Replacement of superconducting shim power supply and magnet power supply circuit breakers \(CB7 and CB8\) on page 50](#)
 - [4.2.10 Replacement of BrainWave and gradient 48V circuit breakers \(CB5 and CB9\) on page 50](#)
 - [4.2.11 Replacement of receptacles and terminal blocks \(REC1-2 and TB2\) on page 51](#)
 - [4.2.12 Replacement of switches and test points \(SW2 and TP1-7\) on page 54](#)
2. Use the schematic as needed to make sure of wire and/or cable placement.
 3. To close the unit back up, follow the [4.2.2 Removal of front panel to access additional FRUs on page 23](#) directions in reverse order.
 4. Once the unit has been closed up, turn on the main input circuit breaker.

4.2.14 Troubleshooting guide

The following table lists fault conditions that may be encountered along with possible causes and solutions.

Module	Symptom	Cause	Solution
PDU	Main input circuit breaker trips	Overtemperature	1. Investigate cause of overtemperature. 2. Allow unit to cool down. 3. Reset the circuit breaker.
		Shorted power off switch	1. Turn power off. 2. Reset the main circuit breaker. 3. Turn power on. 4. If circuit breaker trips, replace the PDU.
		Open from J3-2 to J3-3	1. Disconnect cable at J3. 2. With DVM check for open at cable connector P3-2 to P3-3.
	Distribution breakers tripping	Defective circuit breaker	1. Turn power off. 2. Check if circuit breaker still trips (won't latch up).

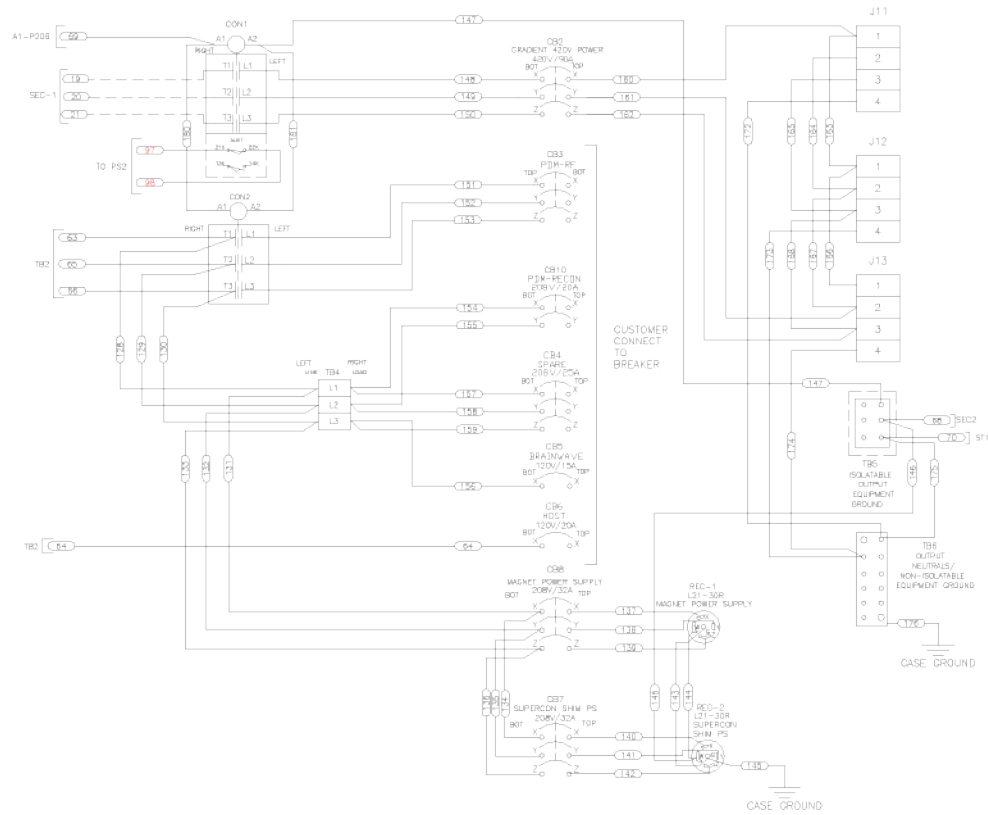
(continued)			
Module	Symptom	Cause	Solution
		Overload	1. Check load by disconnecting cabinet from PDU whose circuit breaker is tripping, or use an Amp Clamp / DVM to check current draw. 2. Correct load problem.
		Defective contactor	1. Make sure that the contactor CON1 is receiving 120V to the coils. 2. If 120V is present, replace the contactor. 3. If 120V is not present, make sure that CON2 is receiving 120V to the coils. 4. If 120V is present, replace the contactor. 5. If 120V is not present, replace the PDU.
	CON1/CON2 contactor not operating	Defective control board A1	1. Disconnect cable at J2. 2. Check at cable connector if P2-7 to P2-8 is shorted. 3. If shorted, replace PDU. 4. If not shorted, connect test plug and verify PDU operation. 5. If PDU operates with test plug, then replace the PDU. 6. If PDU does not operate with test plug, then replace the control board.
		Input circuit breaker off	Turn input circuit breaker, CB1, on.
	Indicators not lighting when power is applied	Input phase missing	1. Check that other phase LEDs are on. 2. Measure at TP.
		Light burned out	1. Check that other phase LEDs are on. 2. Replace PDU.

(continued)			
Module	Symptom	Cause	Solution
		Not getting proper output	1. Measure input and output, at front panel test points, of each phase, from phase to phase (208 VAC: 208 VAC) and phase to ground (120 VAC: 120 VAC). 2. If difference is not a 100:1 ratio, replace the PDU.
	Unit does not exit ESTOP mode (cannot turn on output contactor)	Board not receiving reset signal	Try resetting the unit by pressing SW1 on the front panel.
		Board not receiving power	1. Confirm connection on the board. 2. Confirm board is receiving 120V by measuring at P6-11(hot) and P6-9 (neutral).
	Fans not running	Power supply not receiving power	1. Make sure CB9 is in the ON position. 2. Measure between A1 and A2 at J4 on the PTC board. This should measure +48 VDC whenever the power supply is receiving power.
		Defective power supply	Check the connection to the power supply. If the power supply has good input connection, the power supply is likely defective.
		Defective fan	1. Measure for 48 VDC at the fan connectors P4-1 and P4-3, and P5-1 and P5-3. 2. If 48 VDC is present, then replace the fan.
	Voltage at one or more output missing	Output breaker in the OFF position	Turn ON all output breakers.
		Unit is still in ESTOP mode	Remove ESTOP condition.
	Internal system AC line clock signal missing	Control board failed	Replace control board.
	Power supply LED is off with power applied and PDU is not in ESTOP mode	Defective power supply	Replace power supply.
	Power supply LED is lit but has no 48V output	Defective power supply	Replace power supply.

Chapter 5 Schematic diagrams

5.1 Schematics

Schematic diagrams

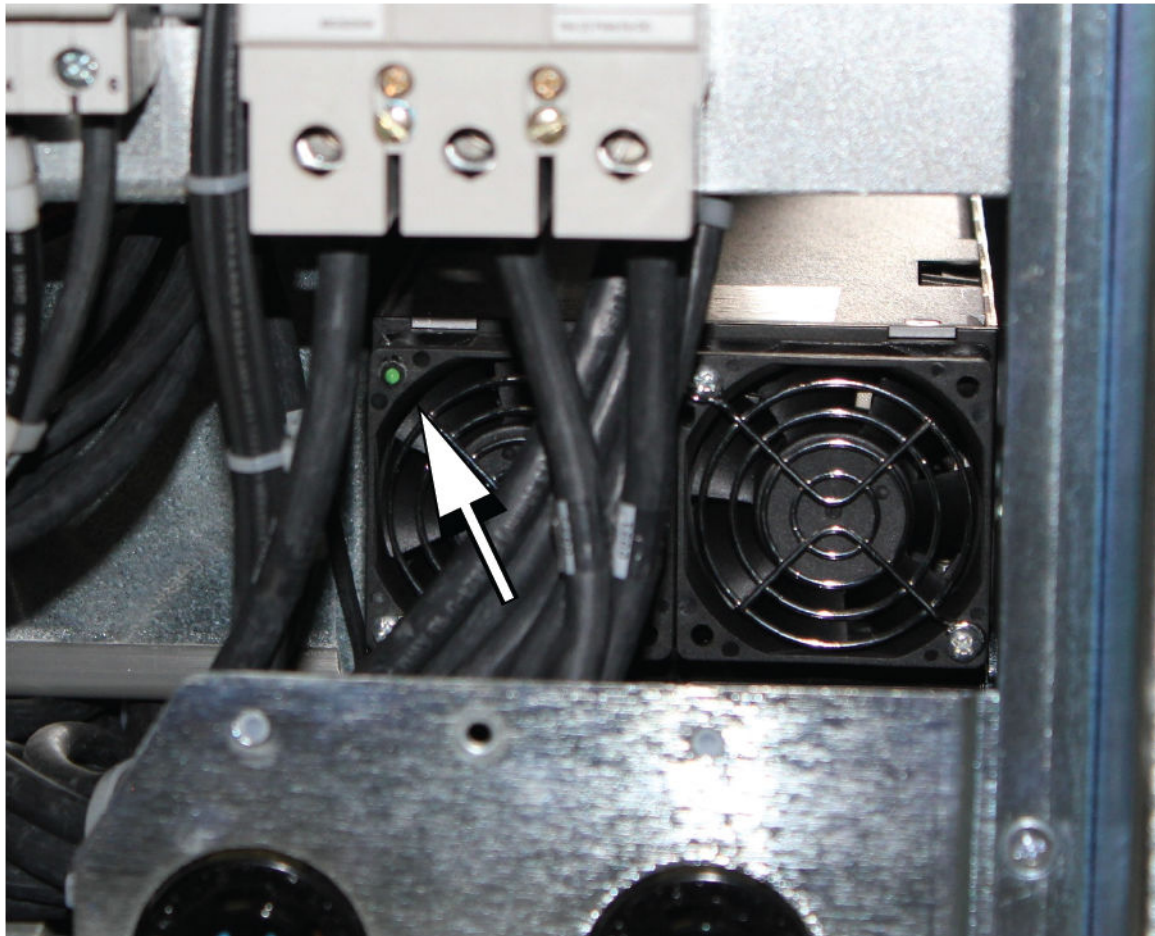


Appendix A Replacement of power supplies

If the power supplies are deemed defective per the [4.2.14 Troubleshooting guide on page 56](#), follow the steps below. The power supplies are mounted behind the receptacle assembly of the PDU located on bottom of the unit.

Each power supply has a green LED in the upper left corner of the leftmost fan. If the LED is off with power applied, then the power supply is most likely defective. If the LED is lit but has no 48V out, then the power supply is most likely defective.

Figure 60 Power On LED

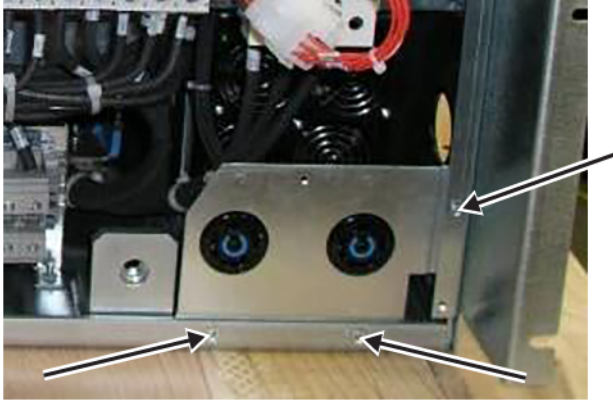


1. Remove the three (10-32 x 1/2) screws at the receptacle assembly and move assembly out of the way.

NOTE

Do not discard the screws, as they will be used to reattach the receptacle assembly.

Figure 61 Receptacle assembly screws



2. If necessary, remove the wiring at the bottom of CB2 and unscrew both circuit breakers from the mount plate for better access to the FRU. Mark wires with masking tape if needed.

Figure 62 Wiring and mounting screws

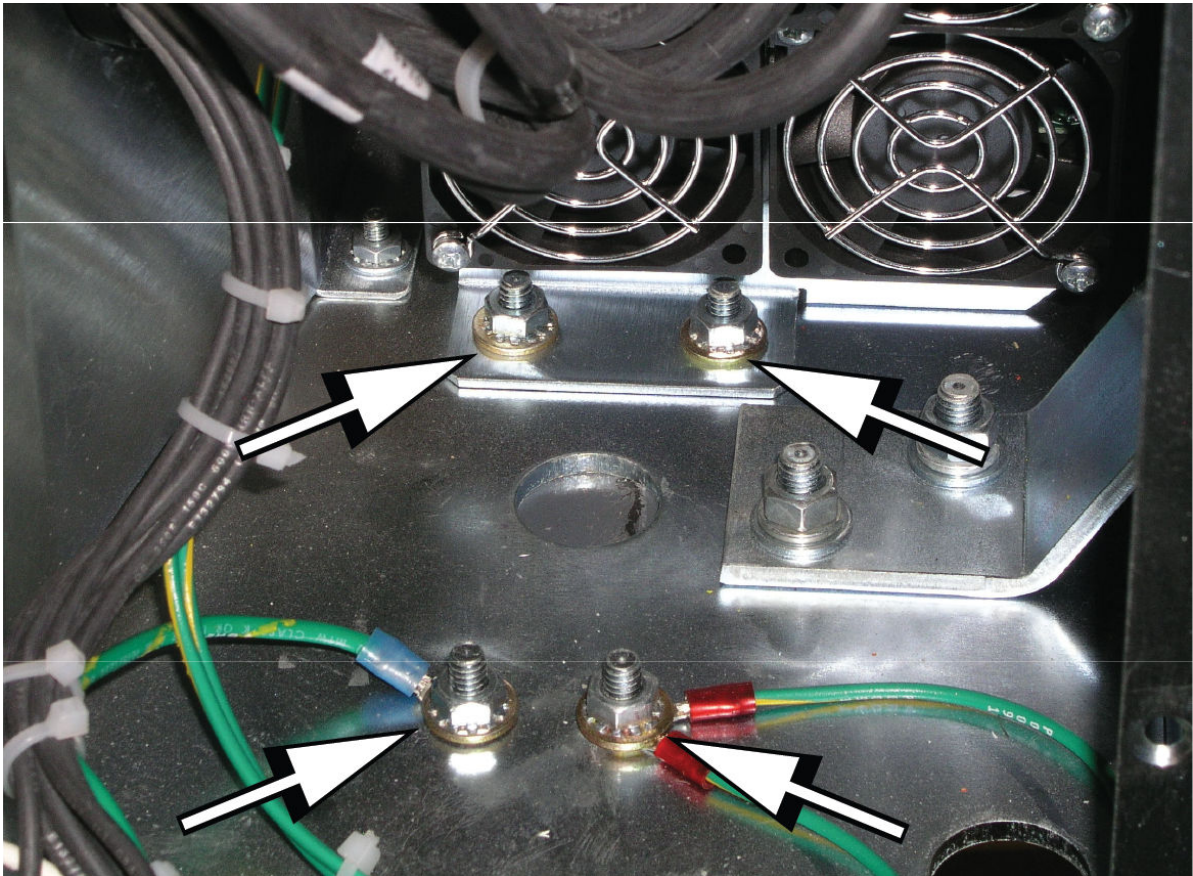


3. Remove two Keps nuts and two washers at the power supply assembly. Remove the two Keps nuts and two washers at the ground wires.

NOTE

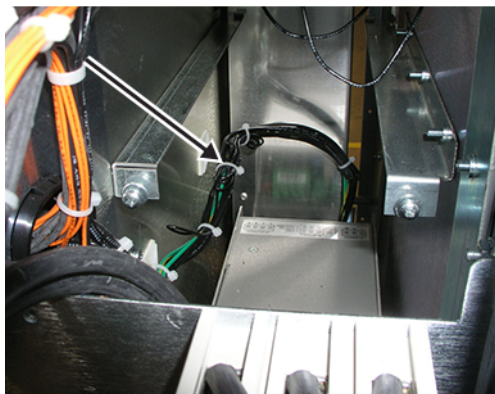
Do not discard the nuts and washers, as they will be used to reattach the power supply assembly and ground wires.

Figure 63 Keps nuts and washers



4. Use small cutter to cut one tie wrap at the power supply harnesses.

Figure 64 Tie wrap



5. Carefully pull out the power supply assembly from the unit. Disconnect all wires from the power supply assembly.

Figure 65 Pulling out the power supply assembly

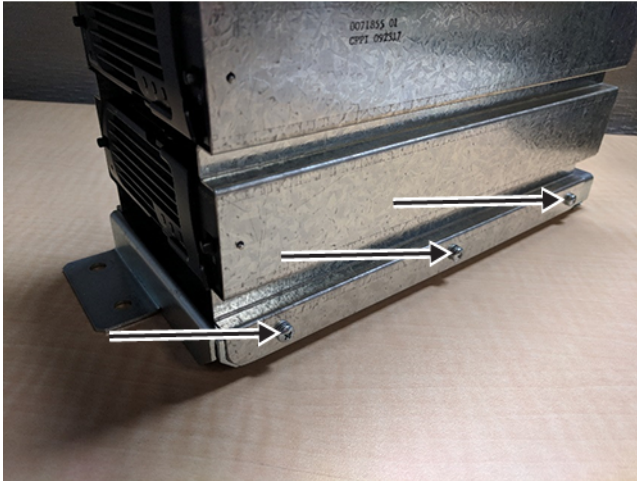


6. Loosen the bottom three screws on one side of the bracket.

NOTE

Do not remove the screws from the bracket.

Figure 66 Bracket screws



7. If replacing power supply PS2, unlatch the mounting pins, disconnect the bottom connector from the rear of the power supply and slide the power supply out of the bracket.

Figure 67 PS2 power supply



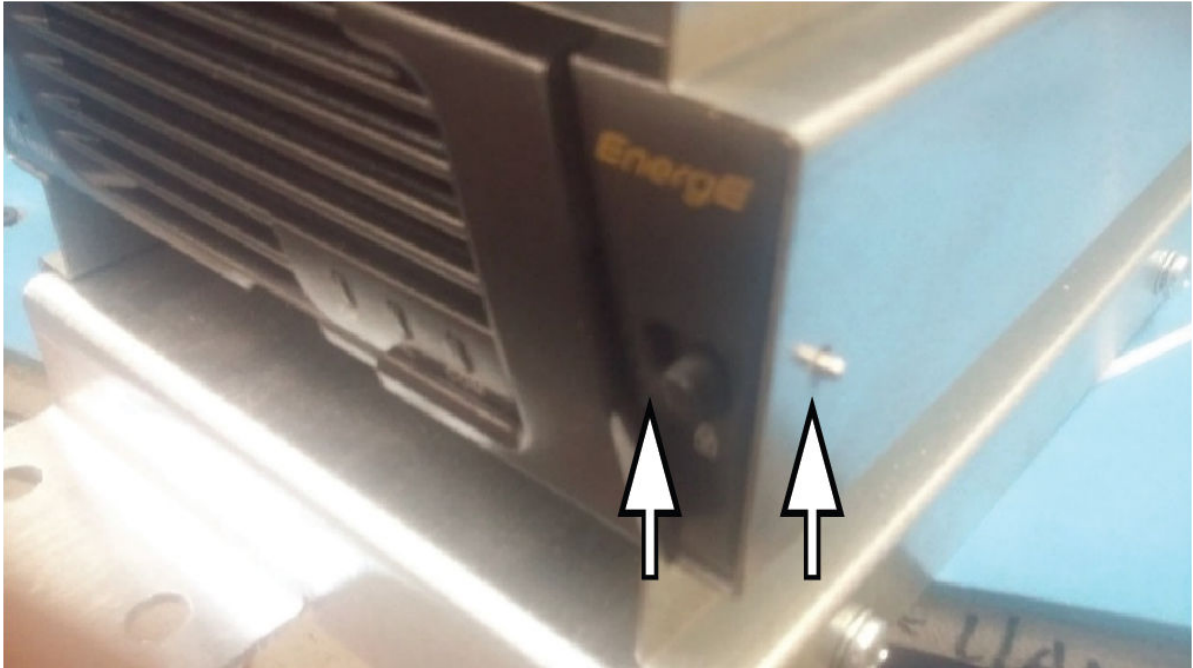
8. Slide the new power supply PS2 into the bottom slot of the bracket and attach the connector on the rear of the power supply.

Figure 68 Inserting PS2 into the bottom slot of the bracket



9. Latch the mounting pins on the power supply through the locating hole on each side of the bracket and slide the button out until the pin extends through the hole.

Figure 69 Mounting pins



10. If replacing power supply PS1, unlatch the mounting pin, disconnect the top connector from the rear of the power supply, and slide the top power supply out of the bracket.

Figure 70 PS1 power supply



11. Slide the new power supply PS1 into the top slot of the bracket and attach the connector on the rear of the power supply.

Figure 71 Sliding PS1 into the top slot of the bracket



12. Latch the mounting pins on the power supply through the locating hole on each side of the bracket (Figure 69 on page 68). See the assembled FRU kit below.

Figure 72 Power supply FRU kit



13. Reinstall the FRU kit assembly by tilting the assembly so that the terminal block clears the CB2 bracket.

Figure 73 Installing the FRU kit assembly



14. Slide the assembly into place where the assembly had been removed. Make sure to move the RF cable out of the way so as not to score the outer jacket with the power supply housing.

Figure 74 Assembly in place



15. Reinstall the mounting hardware (two Keps nuts and two washers) on the power supply assembly bracket and (two Keps nuts and two washers) ground wire connections that were disconnected in [Step 3 on page 64](#).
16. Reinstall CB2 (four screws) if removed in [Step 2 on page 63](#).
17. Reconnect the wires to the bottom of CB2 that were disconnected in [Step 2 on page 63](#).
18. Reinstall the receptacle assembly (three screws) that were removed in [Step 1 on page 61](#).

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TEAL PDU Module (5140621) in XGD Cabinet